SYMMETRY OF VERTICALLY MONOTONE FRACTIONAL ALLEN–CAHN TRAVELING WAVES IN THE PLANE FOR $0 < s \leq \frac{1}{2}$

Abstract. We present a candidate proof of horizontal reflection symmetry and strict horizontal monotonicity for vertically increasing traveling solutions of the planar fractional Allen–Cahn equation, for every $0 < s \leq 1/2$. Only the two pointwise vertical end states are prescribed, and the balanced nondegenerate bistable potential is not required to be even. A renormalized extension stress and a drift-compatible stability argument classify stationary translated limits. A horizontal minimum envelope, a no-merging principle, and rigidity of bounded-width single fronts then give uniform two-arm geometry. Transverse localization and a homogeneous linearized Liouville theorem are used to analyze differences between the wave and its reflection about a frozen midpoint. The corresponding source has width controlled by midpoint displacement rather than interface separation. This yields tail-tight normalized compactness and a finite iteration that makes the midpoint velocity integrable, including at the critical order $s = 1/2$. The limiting midpoint determines the axis in a global moving-plane argument. The auxiliary arguments are developed in sequential order within this manuscript; no earlier working note is an external dependency.

Draft status. This candidate proof was obtained by our team on September 08, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. Introduction

We study vertically increasing traveling profiles for the fractional Allen–Cahn equation in the plane. The transverse coordinate is x and the propagation coordinate is y. Throughout the paper,

$$0 < s \le \frac{1}{2}, \qquad \alpha = 2s \in (0,1], \qquad c > 0, \qquad L_n = (-\Delta_{\mathbb{R}^n})^{\alpha/2}.$$

The operator L_n is normalized to have Fourier symbol $|\xi|^\alpha$. The equation is

$$L_2 u - c u_y + F'(u) = 0 \qquad \text{in } \mathbb{R}^2. \tag{1.1}$$

We assume that $F \in C^{3,\beta}_{\mathrm{loc}}(\mathbb{R})$ for some $0 < \beta < 1$ and

$$\begin{gathered} F(\pm 1) = F'(\pm 1) = 0, \qquad F''(\pm 1) > 0, \qquad F(t) > 0 \quad (-1 < t < 1), \\ F'(t) > 0 \quad (-1 < t < 0), \qquad F'(t) < 0 \quad (0 < t < 1). \end{gathered} \tag{1.2}$$

Thus the two wells are balanced and nondegenerate, with a single interior critical value at 0. The potential may have unequal well curvatures, and no evenness assumption is imposed. The solution satisfies

$$u \in C^2(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2), \qquad |u| < 1, \qquad u_y > 0 \quad \text{in } \mathbb{R}^2, \tag{1.3}$$

and the pointwise limits

$$\lim_{y \to -\infty} u(x,y) = -1, \qquad \lim_{y \to +\infty} u(x,y) = 1 \qquad \text{for every } x \in \mathbb{R}. \tag{1.4}$$

The end-state limits are not assumed uniform in x. In particular, neither lateral phase limits, finite total dissipation, two-arm geometry, a fixed layer trap, nor a limiting midpoint is part of the hypotheses.

For the local Allen–Cahn equation, Gui proved reflection symmetry for monotone traveling waves in the plane with balanced end states [1]. The nonlocal problem introduces a different obstruction at low order. An unrestricted fractional kernel has no first moment when $\alpha \le 1$. Moreover, the absolute extension stress can diverge at large extension height. Even after the two interfaces become locally flat, cancellation of their leading velocities does not by itself imply that their midpoint converges. The argument below uses localized comparison sources rather than an absolute expansion of the full interaction residual.

1.1. **Main statements.**

Theorem 1.1 (Horizontal symmetry)**.** *Let $0 < s \le 1/2$ and $c > 0$. Suppose that (1.1)–(1.4) hold. Then there is a unique $b \in \mathbb{R}$ such that*

$$u(2b - x, y) = u(x, y) \quad \text{for every } (x, y) \in \mathbb{R}^2, \qquad u_x(x, y) < 0 \quad \text{when } x > b. \tag{1.5}$$

Consequently $u_x(x, y) > 0$ for $x < b$.

The geometric conclusions used in the argument are themselves derived from the same assumptions. Let g be the increasing stationary layer, normalized by

$$L_1 g + F'(g) = 0, \qquad g' > 0, \qquad g(0) = 0, \qquad g(-\infty) = -1, \quad g(+\infty) = 1. \tag{1.6}$$

Its existence and uniqueness up to translation follow from the layer theory of Cabré–Sire [3, Theorem 2.4]. Set

$$q = g', \qquad V = F''(g), \qquad M = \int_{\mathbb{R}} q^2 \, dx > 0. \tag{1.7}$$

Theorem 1.2 (Uniform two-arm geometry)**.** *Under the assumptions of Theorem 1.1, the negative vertical limit is uniform and the unique zero graph $y = \gamma(x)$ is proper upward:*

$$\lim_{y \to -\infty} \sup_x |u(x, y) + 1| = 0, \qquad \gamma(x) \longrightarrow +\infty \quad (|x| \to \infty). \tag{1.8}$$

Every horizontally escaping transition sequence, meaning points (x_j, y_j) with $|x_j| \to \infty$ and $|u(x_j, y_j)| \le 1 - \varepsilon$ for some fixed $\varepsilon > 0$, has, after a subsequence, a stationary layer limit under translation, with the corresponding orientation. In particular,

$$u(x_j + X, \gamma(x_j) + T) \longrightarrow \begin{cases} g(-X), & x_j \to +\infty, \\ g(X), & x_j \to -\infty, \end{cases} \quad \text{in } C^2_{\mathrm{loc}}(\mathbb{R}^2), \tag{1.9}$$

and

$$\sup_y u_y(x, y) \longrightarrow 0 \quad (|x| \to \infty). \tag{1.10}$$

For some Y, every slice $y \ge Y$ has exactly two zeros $\ell(y) < r(y)$. They are C^2 and satisfy

$$\begin{gathered} \{x : u(x, y) > 0\} = (\ell(y), r(y)), \qquad \ell' < 0 < r', \qquad \ell', r' \longrightarrow 0, \\ \ell(y) \longrightarrow -\infty, \qquad r(y) \longrightarrow +\infty, \end{gathered} \tag{1.11}$$

with the uniform approximation

$$\|u(\cdot, y) - [g(\cdot - \ell(y)) + g(r(y) - \cdot) - 1]\|_{C^1(\mathbb{R})} \longrightarrow 0. \tag{1.12}$$

The norm in (1.12) concerns the horizontal variable only.

The midpoint is fixed analytically by orthogonality, not by a choice of origin. At large heights there are unique centers a_1, a_2 close to ℓ, r for which, with

$$\Phi(x, y) = g(x - a_1(y)) + g(a_2(y) - x) - 1, \qquad v = u - \Phi,$$

one has $\int vq(x - a_1)\, dx = \int vq(a_2 - x)\, dx = 0$. Write $m = (a_1 + a_2)/2$ and

$$o(x, y) = \frac{u(x + m(y), y) - u(-x + m(y), y)}{2}.$$

Theorem 1.3 (Midpoint convergence before moving planes)**.** *Under the same assumptions, these centers satisfy $a_1 - \ell \to 0$, $a_2 - r \to 0$, and $a_i' \to 0$. There are $b \in \mathbb{R}$, $C < \infty$, and $Y > 1$ such that*

$$|m'(y)| + \|o(\cdot, y)\|_\infty \le C(1 + y)^{-1-2s}, \qquad |m(y) - b| \le C(1 + y)^{-2s} \quad (y \ge Y). \tag{1.13}$$

The number b is the axis in Theorem 1.1.

The estimates in (1.13) are established without assuming symmetry. After symmetry is obtained, uniqueness of the orthogonality coordinates gives $m \equiv b$ and $o \equiv 0$ at large heights. We do not derive a sharp separation law in this paper.

1.2. **Outline and the low-order mechanism.** Section 2 develops analytic resolvent comparison, global regularity, and the layer preliminaries. Section 3 renormalizes the transverse extension stress and proves stability of every stationary translated limit by lifting a compact strict transverse subsolution in a manner compatible with the drift. A half-stress estimate excludes the interior constant; the stationary classification then reduces to the one-dimensional theorem of Cabré–Sire.

Section 4 studies the horizontal infimum envelope. Its one-dimensional balance identity follows by absolutely convergent symmetrization for $\alpha < 1$ and by Hilbert-transform skew-adjointness for $\alpha = 1$. Uniform clearing and a no-merging principle follow. Section 5 combines these facts with a sliding theorem for single fronts of uniformly bounded width. A controlled lateral expansion argument excludes remote defects and yields Theorem 1.2, together with an integrable transverse derivative envelope.

Section 6 proves the homogeneous linearized Liouville theorem in its natural fractional tail class. A source compact in x has a finite localized temporal moment even at low order. This places frequency-filtered solutions in a suitable L^p_x space; a nonzero frequency operator pencil is then invertible. The surviving zero frequency is eliminated up to the translation mode.

In Section 7, the wave is compared with its reflection about a fixed axis b. The part of the comparison equation where the potential is not uniformly positive is supported on sets $S_b(y)$ with

$$|S_b(y)| \le C(1 + |m(y) - b|).$$

The bound is independent of the interface separation. This distinction, and the uniform L^1_x bound for u_x, yield tight fractional tails after normalization on a selected temporal window. Thus the Liouville theorem can be applied to actual nonlinear comparison sequences.

Section 8 proves the growth-to-decay implication

$$|m(y)| = O((1+y)^p) \quad \Longrightarrow \quad |m'(y)| + \|o(\cdot, y)\|_\infty = O((1+y)^{-1-\alpha+p}), \qquad 0 \le p \le 1.$$

Iteration from $p = 1$ reaches an integrable velocity after finitely many steps for each fixed $\alpha > 0$, including the logarithmic borderline. Section 9 uses the finite center to confine all possible moving-plane obstructions and proves Theorem 1.1. The special geometry, stability, and compactness results used in this sequence are proved below, rather than cited to earlier working notes. Only standard layer and boundary regularity results are imported from the published literature [3, 4]; the extension and semigroup frameworks are recalled from [2, 5, 6].

1.3. **Notation.** The symbol C denotes a finite constant whose value may change. Unless a uniformity is explicitly stated, constants may depend on α, c, F and the fixed solution. No estimate is asserted uniformly as $s \downarrow 0$. For $X \in \mathbb{R}^n$, write $\langle X \rangle = (1 + |X|^2)^{1/2}$ and

$$\mathcal{L}_\alpha(\mathbb{R}^n) = \left\{ w : \int_{\mathbb{R}^n} |w(X)| \langle X \rangle^{-n-\alpha} \, dX < \infty \right\}.$$

Our Fourier convention is $\widehat{f}(\xi) = \int_{\mathbb{R}^n} e^{-iX\cdot\xi} f(X)\, dX$, with inverse factor $(2\pi)^{-n}$. The notation $C_b^{k,\theta}$ includes bounds on all derivatives through order k and a global θ-Hölder bound on the derivatives of order k. A stationary limit is independent of the propagation variable y. The extension variable is always $z > 0$.

2. Analytic preliminaries

2.1. **Fractional normalization and positive resolvents.** For sufficiently regular functions in $\mathcal{L}_\alpha$, we use

$$L_n w(X) = C_{n,s}\, \mathrm{PV} \int_{\mathbb{R}^n} \frac{w(X) - w(Z)}{|X - Z|^{n+\alpha}} \, dZ, \qquad C_{n,s} = \frac{4^s \Gamma(n/2 + s)}{\pi^{n/2} |\Gamma(-s)|}. \tag{2.1}$$

The constants are compatible with dimensional marginalization. Let $P_\tau = e^{-\tau L_n}$ have kernel $p^{(n)}_\tau$. Positivity, unit mass, and marginalization follow from Gaussian subordination. We use

the estimates

$$\begin{aligned} |D^j p_\tau^{(n)}(X)| &\le \frac{C_j \tau}{(\tau^{1/\alpha} + |X|)^{n+\alpha+j}} \quad (j = 0, 1, 2), \\ |\partial_\tau p_\tau^{(n)}(X)| &\le \frac{C}{(\tau^{1/\alpha} + |X|)^{n+\alpha}}. \end{aligned} \tag{2.2}$$

The heat-kernel bound $j = 0$ follows from the stable semigroup estimates; see [5, 6]. Differentiating the Gaussian subordination formula gives $\partial_r p_\tau^{(n)}(r) = -2\pi r p_\tau^{(n+2)}(r)$. A second differentiation expresses the Cartesian second derivatives as linear combinations of $p_\tau^{(n+2)}$ and $X_i X_j p_\tau^{(n+4)}$, proving the spatial derivative bounds. Scaling gives $\partial_\tau p_\tau^{(n)} = -(n p_\tau^{(n)} + X \cdot \nabla p_\tau^{(n)})/(\alpha\tau)$, and hence the time-derivative bound. These kernel bounds and the comparison argument below hold for $0 < \alpha < 2$. We use comparison in both dimensions one and two.

Lemma 2.1 (Massive comparison in the tail class)**.** *Let $n \ge 1$, $0 < \alpha < 2$, $\mu > 0$, and $b \in \mathbb{R}^n$. Suppose $e \in \mathcal{L}_\alpha(\mathbb{R}^n)$ is real valued. If*

$$(L_n - b \cdot \nabla + \mu) e \le 0$$

in distributions, then $e \le 0$ almost everywhere. The conclusion is pointwise when e is continuous.

Proof. For $\tau > 0$ set

$$A(\tau, X) = e^{-\mu\tau}(P_\tau e)(X + \tau b).$$

For τ in a compact subinterval of $(0, \infty)$ and X in a compact set, the heat kernel, its spatial derivatives, and its time derivative are bounded by a constant times $\langle Z\rangle^{-n-\alpha}$ as functions of the integration variable Z. Thus all the convolutions below are finite. Testing the distributional inequality gives

$$\partial_\tau A(\tau, X) = -e^{-\mu\tau} P_\tau[(L_n - b \cdot \nabla + \mu)e](X + \tau b) \ge 0. \tag{2.3}$$

Here is the justification for the noncompact test in this formula. Multiply the nonnegative translated heat kernel by $\chi(Z/R)$, where $0 \le \chi \le 1$ is smooth, compactly supported, and identically one near zero. Both this test and its image under the adjoint operator $L_n + b \cdot \nabla + \mu$ are bounded by $C_{X,\tau}\langle Z\rangle^{-n-\alpha}$, uniformly for $R \ge 1$, and converge pointwise to the respective uncut functions. For the fractional term, split increments at $\langle Z\rangle/2$. In the near part, second derivatives of the cutoff heat kernel are bounded by $C\langle Z\rangle^{-n-2-\alpha}$ and symmetric differences apply. In the far part, use its uniformly bounded L^1 norm together with the kernel bound. These estimates give the asserted majorant, also on the cutoff annulus. Dominated convergence against $|e|$ proves (2.3); mollification gives the same argument without initially differentiating e.

For $\tau \ge 1$, the heat-kernel estimate and $\langle Z\rangle \le C\langle X + \tau b\rangle\langle X + \tau b - Z\rangle$ imply

$$|A(\tau, X)| \le C e^{-\mu\tau} \tau \langle X + \tau b\rangle^{n+\alpha} \|e\|_{\mathcal{L}_\alpha} \longrightarrow 0.$$

Hence monotonicity in (2.3) gives $A(\tau, X) \le 0$ for every $\tau > 0$.

Since X is arbitrary, this also gives $P_\tau e \le 0$ everywhere. For $0 \le \varphi \in C_c^\infty(\mathbb{R}^n)$, symmetry of the heat kernel and Fubini's theorem give

$$\int_{\mathbb{R}^n} (P_\tau e)\varphi = \int_{\mathbb{R}^n} e(P_\tau \varphi) \le 0.$$

For $0 < \tau \le 1$, the semigroup L^∞ bound near $\operatorname{supp}\varphi$ and (2.2) away from it imply $|P_\tau\varphi(Z)| \le C_\varphi\langle Z\rangle^{-n-\alpha}$. This bound justifies Fubini and dominated convergence against $|e|$. Since $P_\tau\varphi \to \varphi$ pointwise as $\tau \downarrow 0$, we obtain $\int e\varphi \le 0$. Thus $e \le 0$ almost everywhere, and continuity gives the pointwise assertion. □

For bounded data the same semigroup gives the positive resolvent

$$(L_n - b \cdot \nabla + \mu)^{-1} f(X) = \int_0^\infty e^{-\mu\tau}(P_\tau f)(X + \tau b)\, d\tau. \tag{2.4}$$

Integrating the semigroup derivative against compact tests proves the resolvent equation; the endpoint at infinity vanishes exponentially and the endpoint at zero converges distributionally by the heat-kernel approximate-identity argument used in the comparison proof. Lemma 2.1 gives uniqueness among bounded solutions, applied to real and imaginary parts when necessary. The positive kernel has mass μ^{-1}; consequently its convolution norm on every L^p, $1 \le p \le \infty$, is at most μ^{-1}. In the planar case $b = ce_y$ write $T_\mu = L_2 - c\partial_y + \mu$. In particular, a bounded classical solution of (1.1) satisfies $u = T_1^{-1}(u - F'(u))$.

We shall also use Kato's inequality for distributional solutions. Let $w \in \mathcal{L}_\alpha(\mathbb{R}^n)$ be complex valued and suppose $(L_n - b\cdot\nabla)w = f \in L^1_{\mathrm{loc}}$ in distributions. Set $\operatorname{sgn}_0(\overline{w}) = \overline{w}/|w|$ for $w \neq 0$ and zero otherwise. Then

$$(L_n - b\cdot\nabla)|w| \le \operatorname{Re}\big(\operatorname{sgn}_0(\overline{w})f\big) \quad \text{in distributions.} \tag{2.5}$$

Here is a proof that also covers the nonsmooth functions used below. Mollify with a compactly supported smooth approximate identity, writing $w_\delta = w * \rho_\delta$ and $f_\delta = f * \rho_\delta$. The operator commutes with mollification. For fixed $\varepsilon > 0$, convexity of $\Phi_\varepsilon(z) = \sqrt{|z|^2 + \varepsilon^2} - \varepsilon$ and the chain rule give

$$(L_n - b\cdot\nabla)\Phi_\varepsilon(w_\delta) \le \operatorname{Re}\left(\frac{\overline{w_\delta}}{\sqrt{|w_\delta|^2 + \varepsilon^2}} f_\delta\right).$$

Compact mollification converges locally and in the weighted L^1 norm defining $\mathcal{L}_\alpha$. Since Φ_ε is Lipschitz, the left side therefore converges distributionally as $\delta \downarrow 0$. For the right side use $f_\delta \to f$ in L^1_{loc} and the bounded continuous regularized sign, passing along an almost-everywhere convergent subsequence. Now let $\varepsilon \downarrow 0$. The bounds $0 \le \Phi_\varepsilon(w) \le |w|$ and $|\overline{w}f/\sqrt{|w|^2 + \varepsilon^2}| \le |f|$ justify both limits and prove (2.5).

The same proof applies to $B_\zeta = (-\partial_{xx} + \zeta^2)^{\alpha/2}$ used later. Subordination expresses this operator as a symmetric nonnegative jump operator plus killing at rate $|\zeta|^\alpha$. Convexity handles the jump term; the killing term has the required sign because

$$\Phi_\varepsilon(w) - \frac{|w|^2}{\sqrt{|w|^2 + \varepsilon^2}} = \frac{\varepsilon^2}{\sqrt{|w|^2 + \varepsilon^2}} - \varepsilon \le 0.$$

We use an inhomogeneous Littlewood–Paley decomposition $\sum_{k\ge0} \Delta_k = I$ and write $C^\sigma_* = B^\sigma_{\infty,\infty}$, with norm $\sup_{k\ge0} 2^{k\sigma}\|\Delta_k f\|_\infty$. For $\sigma = k + \theta > 0$ with $k \in \mathbb{N} \cup \{0\}$ and $0 < \theta < 1$, this is $C^{k,\theta}_b$, also denoted by C^σ_b. In particular $L^\infty \subset C^0_*$, and $C^t_* \subset C^\sigma_b$ whenever $0 < \sigma < t$ is noninteger.

Lemma 2.2 (A common resolvent gain)**.** *Let $0 < \alpha \le 1$ and $\mu > 0$. The bounded-data resolvent $R_\mu = T_\mu^{-1}$ admits the dyadic estimate*

$$\|R_\mu f\|_{C^{\sigma+\alpha}_*} \le C_{\alpha,\mu,\sigma}\|f\|_{C^\sigma_*}. \tag{2.6}$$

Here dyadic reconstruction defines $R_\mu f$ for every real σ. If $\sigma + \alpha > 0$, this reconstruction is bounded and satisfies $T_\mu R_\mu f = f$ in distributions. It is then the unique bounded solution of this equation. The constants are independent of the drift c.

Proof. On an annulus $|\xi| \asymp 2^k$, $k \ge 1$, scaling and integration by parts show that a smoothly localized multiplier $e^{-t|\xi|^\alpha}$ has convolution norm at most $Ce^{-at2^{k\alpha}}$. Derivatives on the fixed rescaled annulus introduce only polynomials in $t2^{k\alpha}$, which are absorbed by reducing $a > 0$. The drift only translates this kernel. Consequently, with $\widetilde{\Delta}_k$ a fixed sum of neighboring blocks,

$$\|R_\mu \Delta_k f\|_\infty \le \frac{C}{\mu + 2^{k\alpha}}\|\widetilde{\Delta}_k f\|_\infty.$$

For $k = 0$, convolution with the low-frequency cutoff and the semigroup mass give the bound $C\mu^{-1}\|\widetilde{\Delta}_0 f\|_\infty$. The output of each block has the same Fourier support as its input. Thus $\sum_{k\ge0} R_\mu \Delta_k f$ converges in tempered distributions and satisfies (2.6).

When $\sigma + \alpha > 0$, this series converges uniformly, since its high blocks have norms bounded by $C2^{-k(\sigma+\alpha)}$. Each finite sum solves $T_\mu \sum_{k=0}^N R_\mu \Delta_k f = \sum_{k=0}^N \Delta_k f$, by the bounded-data

semigroup identity. Uniform convergence on the left permits passage against compact tests: the fractional Laplacian of such a test belongs to L^1, as do its drift and mass terms. The right side converges in tempered distributions to f. This proves the distributional equation. Bounded homogeneous uniqueness, already established after (2.4), proves uniqueness here and agreement with that representation for bounded data. □

2.2. **Global regularity from boundedness.**

Lemma 2.3 (Bounded second derivatives without a global regularity assumption)**.** *Let $0 < \alpha \le 1$, $c \in \mathbb{R}$, and $f \in C^2_{\mathrm{loc}}(\mathbb{R})$. Every bounded classical C^2 solution of*

$$(-\Delta)^{\alpha/2}u - c\partial_y u + f(u) = 0 \quad \text{in } \mathbb{R}^2$$

belongs to $C^{2,\gamma}_b(\mathbb{R}^2)$ for every $0 < \gamma < \min(\alpha, 1)$. Its norms are bounded in terms of α, γ, a bound for u, and bounds for f, f', f'' on a compact interval containing its range. The bounds can be chosen independently of c.

Proof. Set $R = T_1^{-1}$ and $H = u - f(u)$. Since u and H are bounded, (2.4) and Lemma 2.1 give

$$u = RH$$

in distributions, before any global derivative bounds are known. We use (2.6), whose constants are independent of c.

For each noninteger $0 < \sigma < 2$, composition with $t \mapsto t - f(t)$ preserves C^σ_b, with its norm controlled by that of u and the stated bounds for f, f', f'' on the range. For $0 < \sigma < 1$ this is the mean-value theorem. For $1 < \sigma < 2$, the identity

$$\nabla H = (1 - f'(u))\nabla u$$

proves the claim: $1 - f'(u)$ is bounded and Lipschitz, with Lipschitz constant at most $\|f''\|_\infty \|\nabla u\|_\infty$, and multiplication by such a coefficient preserves $C^{\sigma-1}_b$.

Fix $0 < \gamma < \alpha$ and put $\sigma_* = 2 + \gamma - \alpha \in (1, 2)$. Choose a finite sequence $0 = \sigma_0 < \sigma_1 < \cdots < \sigma_N = \sigma_*$ whose positive entries are nonintegers and whose increments are strictly smaller than α. Starting from $H \in L^\infty \subset C^0_*$, (2.6) and the preceding composition estimates inductively give $u \in C^{\sigma_k}_b$ at every step: the strict inequality $\sigma_k < \sigma_{k-1} + \alpha$ allows the usual embedding from $C^{\sigma_{k-1}+\alpha}_*$ even when its exponent is an integer. Thus $H \in C^{\sigma_*}_b$. One final application yields

$$u \in C^{\sigma_*+\alpha}_* = C^{2+\gamma}_* = C^{2,\gamma}_b.$$

Every step has bounds in terms of the stated data, and the finite iteration uses no modulus of continuity of f''. Since neither composition nor (2.6) introduces dependence on c, the claimed uniformity follows. □

In particular, the hypotheses of Theorem 1.1 imply global $C^{2,\gamma}_b$ bounds for every fixed $0 < \gamma < \alpha$. Translated solutions are precompact in C^2_{loc}. The fractional operator passes to such bounded limits: symmetric second differences bound increments smaller than δ by $C\delta^{2-\alpha}$, local convergence handles the intermediate annulus, and boundedness gives a uniform $CR^{-\alpha}$ bound outside radius R. In particular, for $j \in \{x, y\}$ the first derivative $w = \partial_j u \in C^{1,\gamma}_b$ satisfies

$$(L_2 - c\partial_y + F''(u))w = 0$$

first in distributions, since constant-coefficient operators commute with differentiation. This equation also holds pointwise. Indeed,

$$|2w(X) - w(X + H) - w(X - H)| \le C|H|^{1+\gamma} \qquad (|H| \le 1),$$

so the symmetric fractional integral over $|H| < \delta$ is bounded uniformly in X by $C\delta^{1+\gamma-\alpha}$. Boundedness of w controls the far integral. The truncated integrals are continuous and converge locally uniformly, identifying their limit with the distribution $L_2 w$. Thus the differentiated equation is classical, with the principal-value interpretation at $\alpha = 1$.

2.3. **Local regularity with a weighted tail bound.**

Lemma 2.4 (Localized constant-drift estimate)**.** *Let $0 < \alpha \le 1$, $c \in \mathbb{R}$, and $R > 0$. Suppose $W \in \mathcal{L}_\alpha(\mathbb{R}^2)$ is locally bounded and solves, in distributions,*

$$(L_2 - c\partial_y + B)W = 0 \quad \text{in } B_{2R}, \qquad B \in C^1(B_{2R}), \quad \|B\|_{C^1(B_{2R})} < \infty.$$

Then W has a $C^{1,\theta}$ representative in B_R for some $\theta > 0$ depending only on α, and there is a constant $C = C(\alpha, c, R, \|B\|_{C^1(B_{2R})})$ such that

$$\|W\|_{C^{1,\theta}(B_R)} \le C\left(\|W\|_{L^\infty(B_{2R})} + \|W\|_{\mathcal{L}_\alpha(\mathbb{R}^2)}\right). \tag{2.7}$$

Consequently, for fixed c, consider families satisfying these hypotheses on every fixed ball for all sufficiently large indices. If they have uniform local L^∞ bounds, uniform weighted-tail bounds, and uniform local C^1 bounds on the coefficients, their regular representatives are precompact in $C^1_{\mathrm{loc}}(\mathbb{R}^2)$. If their weighted tails are also tight, their equations pass to the limit along any subsequence for which the coefficients converge locally uniformly.

Proof. We use the common resolvent gain in Lemma 2.2 with $\mu = 1$, retaining the constant drift in T_1 throughout. The additional ingredient is the commutator with a spatial cutoff.

Commutator estimate. If $\chi \in C_c^\infty$, then

$$[L_2, \chi] : C^\sigma_* \longrightarrow C^{\sigma+1-\alpha}_* \qquad (-1 < \sigma < 1). \tag{2.8}$$

Choose a smooth frequency cutoff ρ equal to one near zero, and split $|\xi|^\alpha = a_{\mathrm{lo}}(\xi) + a_{\mathrm{hi}}(\xi)$, where $a_{\mathrm{lo}} = \rho|\xi|^\alpha$. The low symbol need not be smooth at zero, so its kernel estimate is justified directly. If $K_{\mathrm{lo}} = \mathcal{F}^{-1}a_{\mathrm{lo}}$, Gaussian subordination gives, for every multi-index β,

$$D^\beta K_{\mathrm{lo}} = C_\alpha \int_0^\infty \left[D^\beta \check\rho - e^{t\Delta} D^\beta \check\rho\right] \frac{dt}{t^{1+\alpha/2}}.$$

The integrand has L^1 norm at most

$$\min\{2\|D^\beta \check\rho\|_1, t\|\Delta D^\beta \check\rho\|_1\},$$

so $D^\beta K_{\mathrm{lo}} \in L^1$ for every β. Inserting a fixed low-frequency projector equal to one on $\operatorname{supp}\rho$ therefore shows that $a_{\mathrm{lo}}(D)$ maps every C^σ_* into C^m_b, for each fixed integer m.

For completeness, the multiplication by χ in the first term of $[a_{\mathrm{lo}}(D), \chi]f$ does not discard high input frequencies. Let S be a fixed low output projector, with Schwartz kernel s. For a sufficiently high input block $f_l = \Delta_l f$ and an integer $m \ge 1$, annular localization defines a bounded g_l satisfying

$$f_l = 2^{-2ml}(-\Delta)^m g_l, \qquad \|g_l\|_\infty \le C_m \|f_l\|_\infty.$$

Indeed the rescaled annular multiplier $2^{2ml}|\xi|^{-2m}$ has a uniformly integrable localized kernel. Integrating by parts gives

$$S(\chi f_l)(X) = 2^{-2ml} \int (-\Delta_Z)^m [s(X - Z)\chi(Z)] g_l(Z)\, dZ,$$

and the differentiated bracket has a uniformly bounded L^1_Z norm. Since m is arbitrary, for every N this proves

$$\|S(\chi\Delta_l f)\|_\infty \le C_N 2^{-Nl}\|\Delta_l f\|_\infty.$$

The finitely many remaining input blocks satisfy the ordinary bounded convolution estimate. Choosing $N > |\sigma| + 1$ makes their sum bounded by $C\|f\|_{C^\sigma_*}$. Applying the integrable derivative kernels above gives the same bound for every derivative of $a_{\mathrm{lo}}(D)(\chi f)$. The other term, $\chi a_{\mathrm{lo}}(D)f$, has the same property by the product rule. Thus the low-frequency commutator is smoothing; no smoothness of $|\xi|^\alpha$ at zero is being assumed.

For the high-frequency symbol $a = a_{\mathrm{hi}}$, use

$$[a(D), \chi]f(X) = (2\pi)^{-2} \int \widehat\chi(\vartheta) e^{iX\cdot\vartheta} [a(D + \vartheta) - a(D)] f(X)\, d\vartheta.$$

On a block $|\xi| \asymp 2^l$, when $|\vartheta| \le 2^{l-2}$, the mean-value formula gives

$$|\partial_\xi^\beta [a(\xi+\vartheta) - a(\xi)]| \le C_\beta |\vartheta| 2^{l(\alpha-1-|\beta|)}.$$

Scaling and integration by parts therefore bound its localized convolution norm by $C|\vartheta|2^{l(\alpha-1)}$. To control separated blocks, insert the input and output dyadic cutoffs in the Fourier-shift formula. If $|k-l| > 4$, their supports force $|\vartheta| \ge c_0 2^{\max(k,l)}$. Smoothness and the symbol bounds for a_{hi}, followed by rescaling and a fixed number of integrations by parts, bound the localized convolution norm by a fixed polynomial in $1 + 2^k + 2^l + |\vartheta|$. The Schwartz decay of $\widehat{\chi}$ therefore gives $C_N 2^{-N\max(k,l)}$ for every N. When $|k-l| \le 4$, the region $|\vartheta| \le 2^{l-2}$ gives the preceding mean-value bound, while its complement gives the same rapidly decreasing remainder. Consequently,

$$\|\Delta_k [L_2, \chi] f\|_\infty \le C\, 2^{k(\alpha-1)} \sum_{|l-k|\le 4} \|\Delta_l f\|_\infty + C_N \sum_{l\ge 0} 2^{-N\max(k,l)} \|\Delta_l f\|_\infty.$$

The finitely many low blocks are included by enlarging the constants. Multiplication by $2^{k(\sigma+1-\alpha)}$ and summation of the rapidly decaying terms proves (2.8). The same argument can equivalently be made after regularizing the input, with constants independent of the regularization.

Localization. Fix $\mu = 1$ and choose $\chi, \eta \in C_c^\infty(B_{2R})$, with $\eta = 1$ on a neighborhood of $\operatorname{supp}\chi$. The exact equation is

$$\begin{aligned} T_\mu(\chi W) &= \chi(\mu - B)W - c\chi_y W + [L_2, \chi](\eta W) + \mathcal{R}_{\chi,\eta}, \\ \mathcal{R}_{\chi,\eta}(X) &= C_{2,s}\chi(X) \int \frac{(1-\eta(Z))W(Z)}{|X-Z|^{2+\alpha}}\, dZ. \end{aligned} \tag{2.9}$$

The supports in the last integral are separated. For every integer $j \ge 0$, its C^j norm is bounded by a constant times

$$N_W := \|W\|_{L^\infty(B_{2R})} + \|W\|_{\mathcal{L}_\alpha}.$$

To verify this, differentiate the nonsingular kernel on the bounded integration region and use $\langle Z\rangle^{-2-\alpha}$, or a faster power, on the unbounded region. The cutoff makes $\mathcal{R}_{\chi,\eta}$ compactly supported. Formula (2.9) holds in distributions. At every step below its right side belongs to some C_*^σ with $\sigma + \alpha > 0$. Since χW is bounded, the uniqueness assertion in Lemma 2.2 identifies it with the bounded resolvent reconstruction of that right side. This also specifies the regular representative used below. Representatives on overlapping neighborhoods agree, since they are continuous and coincide almost everywhere.

Here are explicit cutoff and exponent choices. Put $N = \lceil 2/\alpha \rceil$, and choose $\chi_0, \dots, \chi_N \in C_c^\infty(B_{2R})$ with $\chi_{j-1} = 1$ on a neighborhood of $\operatorname{supp}\chi_j$ and $\chi_N = 1$ on a neighborhood of $\overline{B_R}$. Their derivative bounds depend only on R and N. Set

$$\beta_0 = 0, \qquad \beta_j = \min\{j\alpha/2, 1 - \alpha/2\} \quad (1 \le j \le N-1).$$

Initially $\chi_0 W$ is bounded and hence belongs to C_*^0, with norm at most CN_W. Apply the localized equation with $\chi = \chi_j$ and $\eta = \chi_{j-1}$. The nesting of the cutoffs writes the coefficient terms as

$$[\chi_j(\mu - B)](\chi_{j-1}W), \qquad [-c\partial_y \chi_j](\chi_{j-1}W).$$

At the first step its two coefficient terms are bounded directly. At later steps, since $0 < \beta_{j-1} < 1$, multiplication by the compactly supported Lipschitz coefficients $\chi_j(\mu - B)$ and $-c\partial_y\chi_j$ preserves $C_b^{\beta_{j-1}}$. This uses only the given C^1 bound for B; the products are extended by zero outside B_{2R}. The commutator belongs to $C_*^{\beta_{j-1}+1-\alpha} \subset C_*^{\beta_{j-1}}$, and the separated-support remainder is smooth with norm bounded by CN_W. Thus the full right side belongs to $C_*^{\beta_{j-1}}$. The common resolvent estimate and the preceding comparison argument give $\chi_j W \in C_*^{\beta_{j-1}+\alpha}$, with norm at most CN_W. For $j < N$ this implies the retained $C_b^{\beta_j}$ estimate, since $\beta_j < \beta_{j-1} + \alpha$.

Since $\beta_{N-1} = 1 - \alpha/2$, the final step gives

$$\chi_N W \in C_*^{1+\alpha/2} = C_b^{1,\alpha/2}, \qquad \|\chi_N W\|_{C_b^{1,\alpha/2}} \le C N_W.$$

In particular (2.7) holds, for example with $\theta = \alpha/4$. This includes $\alpha = 1$: the initial C_*^1 gain is embedded into $C_b^{1/2}$ before the final step. Only finitely many fixed cutoffs have been used, so all constants have the dependence stated in the lemma.

Arzelà–Ascoli gives compactness of the solutions in C^1_{loc} and, after taking a subsequence, of the coefficients in C^0_{loc}. Here tightness means

$$\lim_{M\to\infty} \sup_j \int_{|Z|>M} |W_j(Z)| \langle Z\rangle^{-2-\alpha}\, dZ = 0.$$

Fatou's lemma gives the same tail bound for the limit. Symmetric differences bound the integral over increments smaller than r by $Cr^{1+\theta-\alpha}$. On a fixed annulus use local convergence. For X in a fixed compact set K and M sufficiently large,

$$\left| \int_{|Z|>M} \frac{W_j(X) - W_j(Z)}{|X-Z|^{2+\alpha}}\, dZ \right| \le C_K \left(M^{-\alpha} \|W_j\|_{L^\infty(K)} + \int_{|Z|>M} |W_j(Z)| \langle Z\rangle^{-2-\alpha}\, dZ \right).$$

The right side tends to zero uniformly in j by local boundedness and tail tightness, and the same estimate holds for the limit by Fatou's lemma. Together with the small-increment and annular estimates, this proves locally uniform convergence of the fractional terms. The drift and zeroth-order terms converge by C^1_{loc} convergence of the solutions and local uniform convergence of the coefficients, so the equation passes to the limit. □

2.4. **The stationary layer and its kernel.** When applying the layer theory, smoothly cut off $-F'$ outside a neighborhood of $[-1,1]$. The resulting nonlinearity is globally $C^{1,\gamma}$ for every $0 < \gamma < 1$, so one may choose $\gamma > \max(0, 1-\alpha)$ as required in [3]. This leaves the equation and potential differences on $[-1,1]$ unchanged. The layer theory [3, Theorems 2.4 and 2.7] gives

$$0 < q(x) \le C\langle x\rangle^{-1-\alpha}, \qquad q \in L^1 \cap L^\infty, \qquad \int_{\mathbb{R}} q = 2, \qquad (L_1 + V)q = 0. \tag{2.10}$$

Since $q, Vq \in L^2$ and $L_1 q = -Vq$ in distributions, Plancherel's theorem gives $q \in H^\alpha$. The layer theory gives a classical C^2 profile. Applying Lemma 2.3 to its planar lift $(x,y) \mapsto g(x)$ with zero drift therefore gives bounded first and second derivatives of g. In particular q' is bounded, and differentiation in distributions gives

$$(L_1 + V)q' = -F'''(g)q^2.$$

Choose $\mu > 0$ and a compact interval I with $V \ge 2\mu$ outside I. The right side of $L_1 q' = -Vq' - F'''(g)q^2$ is bounded, so (2.5) applies directly and yields

$$(L_1 + \mu)|q'| \le Cq^2 + (\mu - V)|q'| \le Cq^2 + C\mathbf{1}_I.$$

This source belongs to $L^1 \cap L^\infty$. Its one-dimensional massive resolvent has the same membership by the kernel mass bound following (2.4). Lemma 2.1 gives

$$|q'| \le (L_1 + \mu)^{-1}(Cq^2 + C\mathbf{1}_I) \in L^1.$$

Thus

$$q, q' \in L^1 \cap L^\infty. \tag{2.11}$$

Lemma 2.5 (The bounded stationary kernel)**.** *For q, V in (1.7), $L_1 + V$ is nonnegative as a quadratic form, and its L^2 kernel is* $\operatorname{span}\{q\}$. *Every bounded classical solution of $(L_1+V)h = 0$ belongs to this same span.*

Proof. We regard $L_1 + V$ as the self-adjoint operator on $L^2(\mathbb{R})$ with domain $H^\alpha(\mathbb{R})$. Since V is real and bounded, its quadratic-form domain is $H^{\alpha/2}(\mathbb{R})$. For a real $\varphi \in C_c^\infty(\mathbb{R})$, the ground-state identity is

$$\langle (L_1 + V)\varphi, \varphi\rangle = \frac{C_{1,s}}{2} \iint \frac{q(x)q(x')}{|x-x'|^{1+\alpha}} \left(\frac{\varphi(x)}{q(x)} - \frac{\varphi(x')}{q(x')} \right)^2 dx\, dx' \ge 0.$$

Indeed, positivity and local C^1 regularity of q make φ^2/q a compactly supported member of $H^{\alpha/2}$. Test $(L_1+V)q=0$ against this function and subtract the resulting bilinear-form identity from the quadratic form of φ. Expansion gives the displayed formula. Its right side is finite: q is bounded below on a neighborhood of $\operatorname{supp}\varphi$, where φ/q is Lipschitz; off that neighborhood the separated kernel and the integrable tail control the integral. Density extends nonnegativity to the form domain, and then to complex functions by real and imaginary parts. Write Q for the full quadratic form of L_1+V. If a real nonzero h lies in the L^2 kernel, then $h\in H^\alpha$ and $Q(h)=0$. Absolute value preserves $H^{\alpha/2}$ and can only decrease the kinetic form, while preserving the potential term. Nonnegativity therefore gives $0\le Q(|h|)\le Q(h)=0$. Since the potential terms agree, the two kinetic forms are equal. Their difference is a positive constant times

$$\iint \frac{|h(x)||h(x')|-h(x)h(x')}{|x-x'|^{1+\alpha}}\,dx\,dx'.$$

The displayed nonnegative integral must vanish; hence h cannot have positive and negative parts of positive measure. If h were not proportional to q, choose a real number t strictly between the essential infimum and essential supremum of h/q. Then $h-tq$ would be a kernel function with positive and negative parts of positive measure, a contradiction. Thus the real kernel is one-dimensional, and so is its complexification.

For a bounded solution h, use the same μ and I and the Kato–resolvent comparison used above for q', now with zero inhomogeneity. It gives

$$|h|\le C(L_1+\mu)^{-1}\mathbf{1}_I\in L^1.$$

Since h is bounded, $h\in L^2$. The identity $L_1h=-Vh\in L^2$ then gives $h\in H^\alpha$ by Plancherel, so it belongs to the operator kernel already identified. □

3. Renormalized stress and stationary translated limits

3.1. **Extension and boundary products.** Let $a=1-\alpha\in[0,1)$ and let $U=\mathcal{P}_z*u$ denote the Caffarelli–Silvestre extension. We use $\mathcal{P}_z$ for the Poisson kernel and P_τ for the heat semigroup. Thus

$$\operatorname{div}(z^a\nabla U)=0,\qquad -\lim_{z\downarrow 0} z^aU_z=d_sL_2u=d_s(cu_y-F'(u)),\tag{3.1}$$

where $d_s>0$ is fixed by this identity. This convention for d_s is the reciprocal of conventions in which $L_2u=d_s\partial_{\nu^a}U$. The Poisson kernel is

$$\mathcal{P}_z^{(n)}(X)=p_{n,s}\frac{z^\alpha}{(z^2+|X|^2)^{(n+\alpha)/2}},\qquad \int_{\mathbb{R}^n}\mathcal{P}_z^{(n)}=1.$$

We use the extension identity and Dirichlet principle from [2]. In particular, for a trace $\xi\in H^{\alpha/2}(\mathbb{R})$,

$$\inf_{\Xi(\cdot,0)=\xi}\int_{\mathbb{R}^2_+} z^a|\nabla\Xi|^2=d_s\langle L_1\xi,\xi\rangle.$$

The pairing on the right is the fractional quadratic form. We shall also use the following continuity of Poisson extension:

$$\begin{aligned} &f_j\to f \text{ locally uniformly},\quad \sup_j\|f_j\|_\infty<\infty\\ \Longrightarrow\quad &\mathcal{P}_z^{(n)}*f_j\to\mathcal{P}_z^{(n)}*f\quad \text{in } C^1_{\mathrm{loc}}(\mathbb{R}^n\times(0,\infty)).\end{aligned}\tag{3.2}$$

Indeed, on any compact positive height interval the Poisson kernel and its first derivatives have uniformly integrable tails. Split each convolution into a fixed ball, where local convergence applies, and its uniformly small tail. This proves the assertion, including the height derivative.

Lemma 3.1 (Uniform conormal bounds and vertical product convergence)**.** *Let $0<\alpha<2$, $a=1-\alpha$, and $u\in C_b^2(\mathbb{R}^2)$. Let $U=\mathcal{P}_z^{(2)}*u$ and let $d_s>0$ be the Dirichlet-to-Neumann normalization for $s=\alpha/2$. With $A_z=z^aU_z$ and $A_0=-d_s(-\Delta)^su$, one has*

$$\sup_{0<z\le 1}\|A_z\|_\infty<\infty,\qquad \|A_z-A_0\|_\infty\longrightarrow 0\quad (z\downarrow 0).$$

Assume in addition that $u_y \geq 0$ and $u(x, \pm\infty) = \pm 1$ for each x. Then, for every compact horizontal interval K,

$$\sup_{x\in K} \int_{\mathbb{R}} |U_y(x,t,z) - u_y(x,t)|\, dt \longrightarrow 0 \quad (z \downarrow 0).$$

Consequently the integrals of $A_z U_y$ over the full vertical line or either fixed vertical half-line converge to the corresponding integrals of $A_0 u_y$, uniformly for $x \in K$.

Proof of 3.1. Write $\mathcal{P}_z(H) = p_\alpha z^\alpha (|H|^2 + z^2)^{-(2+\alpha)/2}$. The kernel

$$K_z(H) = z^{1-\alpha}\partial_z \mathcal{P}_z(H) = p_\alpha \left[\alpha(|H|^2 + z^2)^{-(2+\alpha)/2} - (2+\alpha) z^2 (|H|^2 + z^2)^{-(4+\alpha)/2}\right]$$

is even and has integral zero. Thus

$$A_z(X) = \frac{1}{2} \int K_z(H)[u(X+H) + u(X-H) - 2u(X)]\, dH.$$

The bracket is bounded by $C \min(|H|^2, 1)$, and $|K_z(H)| \leq C|H|^{-2-\alpha}$ for $H \neq 0$. The product of these two majorants is integrable because $0 < \alpha < 2$. Moreover $K_z(H) \to p_\alpha \alpha |H|^{-2-\alpha}$. The same dominated estimate applied to the difference of kernels proves uniform convergence in X to the corresponding fractional-Laplacian integral, which is A_0 by the Dirichlet-to-Neumann normalization. It also proves the asserted uniform bound. The trace u_y is bounded and globally uniformly continuous, so the Poisson approximate identity gives $U_y \to u_y$ uniformly on the plane.

We verify the needed uniform vertical tightness. On a compact horizontal interval the monotone functions $u(x,T)$ increase to 1, and $u(x,-T)$ decrease to -1, uniformly in x by Dini's theorem. Consequently the nonnegative measures $u_y(x,t)dt$, of mass two, have uniform tails there. For $K \subset [-R,R]$, positivity of the Poisson kernel and vertical mass two at every horizontal trace point give

$$\int_{|t|>T} U_y(x,t,z)dt \leq \sup_{|x'|\leq R+1} \int_{|t'|>T-1} u_y(x',t')dt' + 2\int_{|H|>1} \mathcal{P}_z(H)dH.$$

For shifts $|H| \leq 1$, $|t| > T$ implies $|t'| > T-1$ and $|x'| \leq R+1$; on complementary shifts the total vertical mass is two. The last term is $O(z^\alpha)$. Hence the limit superior as $z \downarrow 0$ of the left side tends to zero as $T \to \infty$, uniformly for $x \in K$. On $[-T,T]$ the uniform density convergence gives convergence in $L^1(dt)$. Combining this with the two uniform tail bounds proves the displayed full-line L^1 convergence.

For any one of the asserted integration domains $J \subset \mathbb{R}$,

$$\left| \int_J A_z U_y - A_0 u_y \right| \leq 2\|A_z - A_0\|_\infty + \|A_0\|_\infty \int_{\mathbb{R}} |U_y - u_y| dt.$$

Both terms tend to zero uniformly on the compact horizontal interval. This proves the full-line and half-line product limits. □

3.2. **Renormalized transverse stress.** Set

$$D(x) = \int_{\mathbb{R}} u_y(x,y)^2\, dy, \qquad \Phi_\alpha(R) = \begin{cases} R^{1-\alpha}, & 0 < \alpha < 1, \\ \log(2+R), & \alpha = 1, \end{cases} \quad R \geq 2. \tag{3.3}$$

By monotonicity, $\int u_y(x,y)\, dy = 2$, so the preceding global regularity gives $D(x) \leq 2\|u_y\|_\infty < \infty$.

Lemma 3.2 (Renormalized stress and strip dissipation)**.** *For $Z > 1$ define*

$$H_Z(x) = \int_0^Z \int_{\mathbb{R}} z^a U_x(x,y,z) U_y(x,y,z)\, dy\, dz.$$

For every fixed x_0, the differences $H_Z(x) - H_Z(x_0)$ converge locally uniformly to a C^1 function $\mathscr{H}_{x_0}$ satisfying

$$\mathscr{H}'_{x_0}(x) = -d_s c D(x). \tag{3.4}$$

There is a constant independent of $b \in \mathbb{R}$ and $R \geq 2$ such that

$$\int_{b-R}^{b+R} D(x)\,dx \leq C\Phi_\alpha(R). \tag{3.5}$$

Proof. Poisson positivity and vertical trace mass give

$$U_y \geq 0, \qquad \int_{\mathbb{R}} U_y(x,y,z)\,dy = 2.$$

We establish estimates simultaneously for a fixed integration domain $J = \mathbb{R}$ or either fixed vertical half-line. Write

$$A_J(x,z) = \int_J U_x U_y(x,y,z)\,dy, \qquad H_{J,Z}(x) = \int_0^Z z^a A_J(x,z)\,dz.$$

Thus $H_{\mathbb{R},Z} = H_Z$. All estimates below are independent of the half-line endpoint and of translations of the trace. The Poisson-kernel scaling and the global trace bounds give

$$\|U_x(\cdot,z)\|_\infty \leq C\min(1,z^{-1}), \qquad \|U_{xx}(\cdot,z)\|_\infty \leq Cz^{-2}, \qquad \|U_z(\cdot,z)\|_\infty \leq C/z.$$

The last two estimates are used only for $z \geq 1$. Since $U_{xy} = (\partial_x \mathcal{P}_z) * u_y$, positivity of u_y and its vertical mass give, by Tonelli,

$$\int_{\mathbb{R}} |U_{xy}(x,y,z)|\,dy \leq 2\|\partial_x \mathcal{P}_z\|_1 \leq C/z.$$

Consequently

$$|A_J(x,z)| \leq C\min(1,z^{-1}), \qquad |\partial_x A_J(x,z)| \leq Cz^{-2} \quad (z \geq 1). \tag{3.6}$$

Here differentiation under the improper vertical integral is justified as follows. For fixed $x_*, z > 0$ and $|x - x_*| \leq z/2$, direct comparison of the Poisson kernels gives

$$U_y(x,y,z) + z|U_{xy}(x,y,z)| \leq CU_y(x_*,y,z).$$

The right side is nonnegative and integrable in y. Together with the boundedness of U_x and U_{xx} at fixed z, this dominates $\partial_x(U_xU_y) = U_{xx}U_y + U_xU_{xy}$ by an integrable function uniformly for x near x_*. It proves the asserted derivative formula for both full and half-lines, and its estimate follows from the preceding bounds.

For $L = |x_+ - x_-| \geq 4$, the two bounds in (3.6) imply

$$|A_J(x_+,z) - A_J(x_-,z)| \leq \begin{cases} C, & 0 < z < 1, \\ C\min\{z^{-1}, Lz^{-2}\}, & z \geq 1. \end{cases}$$

In particular the renormalized differences converge locally uniformly as $Z \to \infty$: their integrands are bounded near zero by Cz^a and at infinity by $CLz^{-1-\alpha}$. Splitting the height integral at 1 and L also gives, uniformly for $Z \geq 1$,

$$\begin{aligned} |H_{J,Z}(x_+) - H_{J,Z}(x_-)| &\leq C + C\int_1^L z^{-\alpha}\,dz + CL\int_L^\infty z^{-1-\alpha}\,dz \\ &\leq C\Phi_\alpha(L). \end{aligned} \tag{3.7}$$

For the local convergence assertion, arbitrary bounded endpoint separations are handled by the same near-zero bound and the $CLz^{-1-\alpha}$ tail bound; no lower bound on L is needed there.

The extension stress identity is

$$\partial_x(z^a U_x U_y) = \partial_y\left[\frac{z^a}{2}(U_x^2 - U_y^2 + U_z^2)\right] - \partial_z(z^a U_z U_y). \tag{3.8}$$

Test this identity against a compactly supported smooth function of x and integrate on $-T < y < T$, $\delta < z < Z$. Then let $T \to \infty$. By (1.4) and the translated compactness from Section 2, $u(x, y+t) \to \pm 1$ in C^2_{loc} as $t \to \pm\infty$. Applying (3.2) shows that U_x, U_y, U_z tend to zero uniformly for x in compact sets and $\delta \leq z \leq Z$ at the two vertical ends. This removes the y-boundary terms with the height cutoffs fixed. Lemma 3.1 passes the bottom product to

$$-d_s\int_{\mathbb{R}} (cu_y - F'(u))u_y\,dy = -d_s cD(x) + d_s(F(1) - F(-1)) = -d_s cD(x).$$

The top product is bounded by $CZ^{-\alpha}$. After removing δ,

$$H_Z'(x) = -d_s c D(x) - Z^a \int_{\mathbb{R}} U_z U_y(x,y,Z)\,dy.$$

This equality first holds in distributions. The function D is continuous, since

$$|D(x) - D(x')| \le 4\|u_{xy}\|_\infty |x - x'|,$$

and the top product is continuous by the same Poisson comparison. Integrating in x and letting $Z \to \infty$ proves (3.4).

For $x_- < x_+$ with $x_+ - x_- \ge 4$, the derivative identity gives

$$\mathscr{H}_{x_-}(x_+) = -d_s c \int_{x_-}^{x_+} D(x)\,dx.$$

Since $D \ge 0$ and $c > 0$, (3.7) yields

$$d_s c \int_{x_-}^{x_+} D(x)\,dx = |\mathscr{H}_{x_-}(x_+)| \le C\Phi_\alpha(x_+ - x_-).$$

Taking $x_\pm = b \pm R$ and using $\Phi_\alpha(2R) \le C\Phi_\alpha(R)$ proves (3.5). □

3.3. **Sublinear energy for every stationary translated limit.**

Lemma 3.3 (Half-stress energy transfer)**.** *Suppose $u_j(x,y) = u(x + x_j, y + y_j)$ converges in C^2_{loc} to a stationary function $v = v(x)$. Let $V_0(x,z)$ be the one-dimensional Poisson extension of v. For $R \ge 2$,*

$$\int_{-R}^{R} \left\{ \frac{1}{2} \int_0^\infty z^a ((V_0)_x^2 + (V_0)_z^2)\,dz + d_s F(v(x)) \right\} dx \le C\Phi_\alpha(R). \tag{3.9}$$

No smallness of the dissipation of the translated sequence is required.

Proof. Let $U_j = \mathcal{P}_z * u_j$ and put

$$A_j^-(x,z) = \int_{-\infty}^0 (U_j)_x (U_j)_y(x,y,z)\,dy, \qquad H_{j,Z}^-(x) = \int_0^Z z^a A_j^-(x,z)\,dz.$$

The common flux estimate (3.7), applied to U_j with $J = (-\infty, 0)$ and endpoints $-R, R$, gives

$$|H_{j,Z}^-(R) - H_{j,Z}^-(-R)| \le C\Phi_\alpha(R) \tag{3.10}$$

uniformly in j and $Z \ge 1$.

Fix j, R, Z and integrate (3.8) on $[-R,R] \times (-\infty, 0) \times (\delta, Z)$, first removing a finite lower cutoff in y as in the preceding proof. Then let $\delta \downarrow 0$. Lemma 3.1 gives

$$\int_{-\infty}^0 z^a (U_j)_z (U_j)_y\,dy \longrightarrow -d_s c \int_{-\infty}^0 (u_j)_y^2\,dy + d_s F(u_j(x,0)).$$

This uses the original end state $u_j(x,-\infty) = -1$ for each fixed j; no such end state is imposed on the limiting function. Define

$$E_{j,R,Z} = \int_{-R}^{R} \left\{ \frac{1}{2} \int_0^Z z^a ((U_j)_x^2 + (U_j)_z^2)(x,0,z)\,dz + d_s F(u_j(x,0)) \right\} dx.$$

Write $E_{R,Z}(v)$ for the same expression with U_j replaced by V_0 and $u_j(x,0)$ by $v(x)$. After the bottom cutoff is removed, the exact half-stress identity is

$$\begin{aligned} E_{j,R,Z} = {} & H_{j,Z}^-(R) - H_{j,Z}^-(-R) + \frac{1}{2} \int_{-R}^R \int_0^Z z^a (U_j)_y(x,0,z)^2\,dz\,dx \\ & + d_s c \int_{-R}^R \int_{-\infty}^0 (u_j)_y^2\,dy\,dx + Z^a \int_{-R}^R \int_{-\infty}^0 (U_j)_z (U_j)_y(x,y,Z)\,dy\,dx. \end{aligned}$$

The positive z-energy is integrable down to zero: the uniform conormal bound gives $z^a (U_j)_z^2 \le C z^{-a}$ there, while bounded tangential derivatives give $z^a (U_j)_x^2 \le C z^a$; both

powers are integrable because $-1 < a < 1$. The last term has absolute value at most $CRZ^{-\alpha}$ by $\|(U_j)_z(\cdot, Z)\|_\infty \le C/Z$ and vertical mass two. Thus (3.10) yields

$$E_{j,R,Z} \le C\Phi_\alpha(R) + \frac{1}{2}\int_{-R}^{R}\int_0^Z z^a (U_j)_y(x,0,z)^2\, dz\, dx + d_s c \int_{-R}^{R}\int_{\mathbb{R}} (u_j)_y^2\, dy\, dx + CRZ^{-\alpha}. \tag{3.11}$$

The strip estimate (3.5) is invariant under both translations, so the full-line dissipation term in (3.11) is at most $C\Phi_\alpha(R)$, uniformly in j.

Only now let $j \to \infty$, keeping R, Z fixed. By (3.2) and kernel marginalization, $U_j \to V_0(x,z)$ with first derivatives at positive heights; in particular $(U_j)_y \to 0$. The uniform bound $|(U_j)_y| \le \|u_y\|_\infty$ permits dominated convergence down to $z = 0$. Fatou's lemma for the nonnegative derivative energy and convergence of the potential now give

$$E_{R,Z}(v) \le C\Phi_\alpha(R) + CRZ^{-\alpha}.$$

Let $Z \to \infty$ and use monotone convergence to obtain (3.9). Since $|v| \le 1$ and $F \ge 0$ on $[-1,1]$, discarding the nonnegative derivative energy gives $0 \le \int_{-R}^{R} F(v) \le C\Phi_\alpha(R) = o(R)$. □

3.4. **Stability survives the vertical drift.**

Definition 3.4 (Transverse stability)**.** For a bounded real coefficient $B = B(x)$, define

$$Q_B(\xi) = \frac{C_{1,s}}{2}\iint_{\mathbb{R}^2} \frac{|\xi(x) - \xi(x')|^2}{|x - x'|^{1+\alpha}}\, dx\, dx' + \int_{\mathbb{R}} B(x)\xi(x)^2\, dx.$$

The operator $L_1 + B$ is stable if $Q_B(\xi) \ge 0$ for every real $\xi \in C_c^\infty(\mathbb{R})$. The equation, potential, and normalization in this definition are those of (1.1)–(1.2) and (2.1).

A positive vertical derivative cannot simply be declared to be a positive solution of the drift-free transverse linearized equation. The next three lemmas supply the needed transfer. In these lemmas the compact transverse test is denoted by ζ; it is distinct from the layer derivative q in (1.7).

Lemma 3.5 (A negative quadratic test yields a compact strict subsolution)**.** *Use the operator and quadratic form of Definition 3.4. Let $b \in C_b^1(\mathbb{R})$ and let $D = (a_-, a_+)$ be a bounded interval. Suppose $Q_b(\xi_0) < 0$ for some nonzero $\xi_0 \in C_c^\infty(D)$. Then there is a nonzero nonnegative globally Lipschitz function ζ, compactly supported in D, and a number $\delta_0 > 0$ such that*

$$L_1\zeta(x) + b(x)\zeta(x) \le -\delta_0 \qquad (\zeta(x) > 0).$$

On its positive set ζ has local C^γ regularity for some $\gamma > \alpha$, so the displayed fractional operator exists pointwise.

Proof of 3.5. Let $H_0^s(D)$ denote the closure of $C_c^\infty(D)$ in $H^s(\mathbb{R})$, with zero exterior extension. For an interval this is precisely the space of $H^s(\mathbb{R})$ functions vanishing almost everywhere on D^c: inward dilation about the midpoint of D moves the support strictly inside D and converges strongly in $H^s(\mathbb{R})$ as its factor tends to one; subsequent mollification gives approximants in $C_c^\infty(D)$. This argument also covers $s = 1/2$. On $H_0^s(D) \cap L^4(D)$ minimize

$$J(w) = \frac{1}{2}\langle L_1 w, w\rangle + \frac{1}{2}\int_D b(x)w^2\, dx + \frac{1}{4}\int_D w^4\, dx.$$

Here $\langle L_1 w, w\rangle$ is the nonnegative double integral in Definition 3.4. The pointwise inequality $b(x)t^2/2 + t^4/4 \ge t^4/8 - C\|b_-\|_\infty^2$ gives a lower bound for J and bounds both the kinetic seminorm and the L^4 norm on any minimizing sequence. Fractional Poincaré on the bounded interval bounds the full H^s norm. A subsequence converges weakly in H_0^s and L^4, strongly in L^2 by compact embedding, and almost everywhere. The potential quadratic term converges strongly; the kinetic and quartic terms are weakly lower semicontinuous. Thus J has a minimizer. For small $t > 0$,

$$J(t\xi_0) = \frac{t^2}{2}Q_b(\xi_0) + \frac{t^4}{4}\int_D \xi_0^4 < 0,$$

so the minimizer is nonzero.

Absolute value and clipping at a positive level are 1-Lipschitz contractions vanishing at zero; they preserve $H_0^s(D)$ and decrease the fractional seminorm. Replacing the minimizer by its absolute value therefore decreases the kinetic term and preserves the other terms. Clipping that nonnegative minimizer at $M \geq \max\{1, \|b_-\|_\infty^{1/2}\}$ also decreases the kinetic term. The pointwise potential is nondecreasing for $t \geq M$, since its derivative is $t(b(x)+t^2) \geq 0$ there. We may consequently choose a nonzero minimizer ϕ with $0 \leq \phi \leq M$. It is still an unconstrained global minimizer, so compact signed variations give

$$L_1\phi + b(x)\phi + \phi^3 = 0 \text{ weakly in } D, \qquad \phi = 0 \text{ outside } D.$$

After the bounded minimizer has been chosen, the reaction $-b\phi - \phi^3$ is bounded. Thus the identity for compact smooth tests extends by density to all $H_0^s(D)$ tests, which is the weak formulation used in the boundary regularity theorem.

Apply [4, Corollary 1.6(a)] with reaction $(x,t) \mapsto -b(x)t - t^3$. The interval is a bounded $C^{1,1}$ domain, the reaction is Lipschitz on $\overline{D} \times [-M, M]$, and ϕ is a bounded weak solution with zero exterior data. The cited result gives $\phi \in C^s(\mathbb{R})$ and interior C^γ regularity for every noninteger $\gamma < 1 + \alpha$. Choose $1 < \gamma < 1 + \alpha$. In particular the zero exterior extension is continuous, ϕ is continuously differentiable in D, and $L_1\phi$ exists pointwise and is continuous there. Its weak equation therefore holds pointwise. If ϕ vanished at an interior point, its fractional Laplacian would be strictly negative there because $\phi \geq 0$ is nonzero, whereas the equation would give zero. Thus $\phi > 0$ throughout D.

Put $B = \|b\|_\infty$. Choose a small collar width $\rho > 0$, less than one third the interval length, so that $(C_{1,s}/\alpha)\rho^{-\alpha} > B + 1$. On the compact middle interval $[a_- + \rho, a_+ - \rho]$, positivity gives a minimum $m > 0$. Choose $\varepsilon > 0$ so small that $\varepsilon < m/2$ and $B\varepsilon \leq m^3/2$, and put $\zeta = (\phi - \varepsilon)_+$. Its support is compactly contained in D, because the continuous exterior trace of ϕ is zero. Interior regularity makes its derivative bounded on $\{\phi \geq \varepsilon\}$, so this clipped function is globally Lipschitz. At every point x where $\zeta(x) > 0$, the exact identity is

$$L_1\zeta(x) - L_1\phi(x) = -C_{1,s} \int_{\mathbb{R}} \frac{\zeta(t) - \phi(t) + \varepsilon}{|x - t|^{1+\alpha}}\, dt.$$

The numerator is nonnegative everywhere and vanishes in a neighborhood of this x. On D^c it is exactly ε. Writing $\delta(x) = \operatorname{dist}(x, D^c)$ and integrating over the nearer exterior half-line gives

$$L_1\zeta - L_1\phi \leq -\frac{C_{1,s}}{\alpha}\varepsilon\delta(x)^{-\alpha}.$$

Consequently

$$(L_1 + b)\zeta \leq -\phi^3 - b\varepsilon - \frac{C_{1,s}}{\alpha}\varepsilon\delta(x)^{-\alpha}.$$

If $\delta(x) < \rho$, the right side is at most $-\varepsilon$ by the choice of ρ. On the middle interval it is at most $-m^3/2$ by the choice of ε. Taking the smaller positive constant proves the uniform strict margin. The local regularity on the positive set is the same as that of ϕ; no fractional equation is evaluated at a clipping corner. No sign assumption on b near the interval endpoints has been used. □

Lemma 3.6 (A compact strict transverse subsolution survives vertical drift)**.** *Use Definition 3.4. Suppose b is bounded and $\zeta \geq 0$ is nonzero and compactly supported in a fixed bounded interval I, is bounded and continuous globally, and has local C^γ regularity with $\gamma > \alpha$ wherever it is positive. Suppose*

$$L_1\zeta + b\zeta \leq -\delta_0 < 0 \quad \text{on } \{\zeta > 0\}.$$

Then there is a nonnegative compactly supported continuous function ψ on $\mathbb{R}^2$ and $\nu > 0$ such that $\psi \not\equiv 0$ and

$$(L_2 - c\partial_y + b(x))\psi \leq -\nu \quad \text{on } \{\psi > 0\}.$$

The derivative $\partial_y\psi$ exists continuously, and $L_2\psi$ exists pointwise on its positive set.

Proof of 3.6. Let $\eta(Y) = \exp[-1/(1-Y^2)]$ for $|Y| < 1$, extended by zero, and set $\psi_T(x,y) = \zeta(x)\eta(y/T)$, where $T > 1$ and $Y = y/T$. Kernel marginalization gives

$$(L_2 - c\partial_y + b)\psi_T = \eta(Y)(L_1 + b)\zeta + E_T(x,y) - \frac{c}{T}\zeta(x)\eta'(Y),$$

where

$$E_T(x,y) = \frac{C_{2,s}}{2}\iint_{\mathbb{R}^2} \frac{\zeta(x')S(Y,t/T)}{((x-x')^2+t^2)^{1+\alpha/2}}\,dx'\,dt,$$
$$S(Y,r) = 2\eta(Y) - \eta(Y+r) - \eta(Y-r).$$

To justify the decomposition at a positive point, first use a circular principal-value cutoff. Pair the vertical increments in the error term; their second difference is $O(t^2)$, so this term is absolutely integrable. Pair the horizontal increments in the remaining term; local C^γ regularity with $\gamma > \alpha$ makes those symmetric differences integrable against the one-dimensional marginal. The cutoff can therefore be removed. This also proves pointwise existence of $L_2\psi_T$ on its positive set.

Let E_T^+ denote the same integral as E_T with S replaced by $S_+ = \max\{S, 0\}$. Since $\zeta \geq 0$,

$$\int_{\mathbb{R}} \frac{\zeta(x')}{((x-x')^2+t^2)^{1+\alpha/2}}\,dx' \leq C\|\zeta\|_\infty |t|^{-1-\alpha}.$$

For $d = 1 - |Y| > 0$, differentiation of $\log\eta$ gives

$$|\eta'(Y)| \leq C\eta(Y)d^{-2}, \qquad \sup_{|r|\leq c_0 d^2} |\eta''(Y+r)| \leq C\eta(Y)d^{-4}, \tag{3.12}$$

where $c_0 > 0$ is a fixed sufficiently small constant. Indeed, on this interval the distance to the cutoff edge is comparable to d, $\log\eta$ varies by a bounded amount, and its first two derivatives are respectively $O(d^{-2})$ and $O(d^{-3})$. Use $S_+ \leq C\eta(Y)d^{-4}r^2$ for $|r| \leq c_0 d^2$, and $S_+ \leq 2\eta(Y)$ otherwise. Integrating the preceding marginal yields

$$E_T^+(x,y) \leq C\eta(Y)(Td^2)^{-\alpha}, \qquad \left|\frac{c}{T}\zeta(x)\eta'(Y)\right| \leq C\eta(Y)(Td^2)^{-1}. \tag{3.13}$$

Thus, when $d \geq T^{-1/4}$, both error terms are $o(\eta(Y))$ uniformly. The strict transverse inequality gives, for large T,

$$(L_2 - c\partial_y + b)\psi_T \leq -\tfrac{1}{2}\delta_0\eta(Y) \quad \text{if } \zeta(x) > 0,\ d \geq T^{-1/4}. \tag{3.14}$$

On the remaining region $0 < d < T^{-1/4}$, use $\eta(Y) \leq e^{-1/(2d)}$. For each fixed $k \geq 0$, $d^{-k}e^{-1/(2d)}$ is increasing for sufficiently small d. Consequently (3.13) implies

$$\eta(Y) \leq e^{-c_1 T^{1/4}},$$
$$E_T^+(x,y) + \left|\frac{c}{T}\zeta(x)\eta'(Y)\right| \leq C(T^{-\alpha/2} + T^{-1/2})e^{-c_1T^{1/4}} \tag{3.15}$$
$$= o(T^{-1-\alpha}).$$

There is a uniformly negative contribution to E_T in this region. For $Y > 0$, which is at least $3/4$ once T is large, restrict to $Y' \in [-1/4, 1/4]$ and $t = T(Y' - Y)$. Then $\eta(Y') \geq \eta_* > 0$ and $\eta(2Y - Y') = 0$; hence $S(Y, t/T) \leq -\eta_*/2$ for large T. For x in the positive set of ζ and x' in its support, $|x - x'| \leq \operatorname{diam} I$, while $|t|$ is comparable to T. The kernel is therefore bounded below by $cT^{-2-\alpha}$. Integration over this vertical interval of length $T/2$ and over x' gives a negative contribution at most $-c_2T^{-1-\alpha}\int\zeta$, uniformly in x. Absorb the fixed positive mass $\int\zeta$ into c_2. For $Y < 0$ the same choice of central interval gives the identical bound. Thus $E_T \leq E_T^+ - c_2T^{-1-\alpha}$, and (3.15), together with the nonpositive transverse term, gives

$$(L_2 - c\partial_y + b)\psi_T \leq -\tfrac{1}{2}c_2T^{-1-\alpha} \quad \text{if } \zeta(x) > 0,\ 0 < d < T^{-1/4}. \tag{3.16}$$

Fix one sufficiently large T. The minimum of η on $|Y| \leq 1 - T^{-1/4}$ is positive, so the smaller of the margins in (3.14) and (3.16) is a number $\nu > 0$. The function $\psi = \psi_T$ is nonzero, nonnegative, continuous, and compactly supported, and $\partial_y\psi$ is continuous globally. This proves all the assertions. □

Lemma 3.7 (Stationary translated limits inherit stability from positive vertical differences)**.** *Use Definition 3.4. Let $u_j \in C_b^2(\mathbb{R}^2)$ be bounded classical solutions of $L_2 u_j - c(u_j)_y + F'(u_j) = 0$, with $|u_j| < 1$ and $(u_j)_y > 0$. Suppose $u_j \to v(x)$ locally uniformly, where $v \in C_b^2(\mathbb{R})$ is stationary and $|v| \le 1$. Then $L_1 + F''(v)$ is stable in the sense of Definition 3.4.*

Proof of 3.7. Put $b(x) = F''(v(x))$. The potential and derivative assumptions in Definition 3.4 and $v \in C_b^2$ give $b \in C_b^1$. Suppose some $\xi_0 \in C_c^\infty(\mathbb{R})$ had $Q_b(\xi_0) < 0$. Choose any bounded interval D containing its support. Apply Lemma 3.5 to obtain ζ with a compact strict negative residual. Then Lemma 3.6 gives a fixed compact nonnegative function ψ and $\nu > 0$ with

$$(L_2 - c\partial_y + b(x))\psi \le -\nu \text{ on } \{\psi > 0\}.$$

Fix a vertical increment $\tau > 0$ and put $h_j(x,y) = u_j(x, y+\tau) - u_j(x,y) > 0$. Subtraction of the two wave equations gives, pointwise,

$$(L_2 - c\partial_y + b_j(x,y))h_j = 0, \quad b_j(x,y) = \int_0^1 F''(u_j(x,y) + \theta h_j(x,y))\, d\theta.$$

For this fixed τ, stationarity of the local limit implies $h_j \to 0$ locally and $b_j \to b$ locally uniformly. The test ψ is fixed and bounded, so for all large j,

$$(L_2 - c\partial_y + b_j)\psi \le -\nu/2 \quad \text{on } \{\psi > 0\}.$$

Each h_j is continuous and strictly positive on the compact support of ψ, and is bounded on the full space. Therefore $t_j = \inf_{\psi>0} h_j/\psi$ is a finite positive number attained where $\psi > 0$: the ratio tends to infinity at the boundary of its positive set because h_j has a positive compact minimum and ψ vanishes there. The function $w = h_j - t_j\psi$ is globally nonnegative and vanishes at such a contact point. At that point its fractional Laplacian is nonpositive, its y derivative is zero, and $b_j w = 0$. The local regularity in Lemma 3.6 suffices to evaluate the fractional operator there; clipping corners lie outside the positive set. On the other hand,

$$(L_2 - c\partial_y + b_j)w = -t_j(L_2 - c\partial_y + b_j)\psi > 0,$$

a contradiction. Thus no negative quadratic test exists. The argument uses positive finite differences, and makes no assumption about a normalized limit of $(u_j)_y$ or about uniform positive lower bounds for h_j as $j \to \infty$. □

3.5. **Classification and the translated orbit.**

Proposition 3.8 (Sublinear-energy stationary translated limits are pure phases or layers)**.** *Under the hypotheses of Lemma 3.7, suppose also $\int_{-R}^{R} F(v) = o(R)$. Let g denote the unique increasing solution of*

$$L_1 g + F'(g) = 0, \qquad g(\pm\infty) = \pm 1, \qquad g(0) = 0.$$

Then either $v \equiv -1$, $v \equiv 1$, or, for some $b \in \mathbb{R}$,

$$v(x) = g(x - b) \quad \text{or} \quad v(x) = g(b - x).$$

If additionally $v(0) = 0$, only the last two alternatives occur and $b = 0$.

Proof of 3.8. Lemma 2.3 and the compactness and equation-passage consequences established in Section 2 upgrade the prescribed convergence to C^2_{loc} and pass the equation to v. Since v is independent of y, kernel marginalization gives $L_1 v + F'(v) = 0$.

Use Definition 3.4 and the stability from Lemma 3.7. If v is constant, its stationary equation gives $F'(v) = 0$. The strict signs of F' make $-1, 0, 1$ its only possible values in $[-1, 1]$. The value zero is excluded by the sublinear potential integral, since $F(0) > 0$; the two pure phases are exactly the constant alternatives in the conclusion. Suppose henceforth that v is nonconstant. The maximum principle excludes attained values ± 1: at an attained endpoint, the stationary fractional equation with $F'(\pm 1) = 0$ would force the bounded profile to be that constant. Thus $|v| < 1$.

Apply [3, Theorem 2.9(c)]. We check the exact extension normalization and hypotheses. Let V be the extension of v and normalize $\partial_{\nu^a} V = d_s L_1 v$, $a = 1 - \alpha$. Their boundary equation is $(1+a)\partial_{\nu^a} V = f_{\mathrm{ext}}(V)$, so on a neighborhood of $[-1,1]$ take

$$f_{\mathrm{ext}} = -(1+a) d_s F', \qquad G_{\mathrm{ext}} = (1+a) d_s F.$$

This is a positive scaling of the same potential. Their required $f_{\mathrm{ext}} \in C^{1,\gamma}$ with $\gamma > \max(0, 1-\alpha)$ follows from the cutoff construction for the layer theory in Section 2, followed by the positive scaling above. Choose the potential primitive to agree with G_{ext} on $[-1,1]$. The strict signs of F' imply that the only critical points in $[-1,1]$ are $-1, 0, 1$. Since $F(0) > F(\pm 1) = 0$, the only distinct equal-potential critical pair is $-1, 1$. Thus their condition (2.13)–(2.15) holds, and remains true after the positive scaling above.

The positive Poisson kernel gives $|V| < 1$. Its weighted energy is finite on bounded half-balls: tangential derivatives are bounded, and the conormal bound in Lemma 3.1, applied to the planar lift of v and then marginalized, gives $z^a V_z^2 \le C z^{-a}$ near zero. Since $-1 < a < 1$, both derivative energies are locally integrable. The extension equation and its conormal identity therefore hold in the weak sense by integration above $z = \delta$ and passage to zero. This places V in the weak extension class of the cited theorem.

Because $F''(v)$ is bounded, density extends trace stability from $C_c^\infty(\mathbb{R})$ to $H^{\alpha/2}(\mathbb{R})$. For any compactly supported C^1 extension test Ξ up to $z = 0$, its trace ξ belongs to this space. The Dirichlet principle gives

$$\int z^a |\nabla \Xi|^2 \, dx \, dz + d_s \int F''(v) \xi^2 \, dx \ge d_s Q_{F''(v)}(\xi) \ge 0.$$

Their potential coefficient is exactly $-f'_{\mathrm{ext}}/(1+a) = d_s F''$, so this is the required inequality. The theorem therefore makes v a strictly monotone layer, with one of the two orientations. Apply the uniqueness modulo translation of the increasing layer g already fixed in (1.6), reflecting x in the decreasing case. This gives $v(x) = g(x-b)$ or $v(x) = g(b-x)$ for some $b \in \mathbb{R}$. If $v(0) = 0$, the pure phases are impossible and strict increase of g fixes the remaining translation parameter to zero. □

Combining Lemma 3.3, Lemma 3.7, and Proposition 3.8 gives the following conclusion under the original assumptions:

$$\text{every stationary translated limit is} \quad -1, \quad 1, \quad g(x-b), \quad \text{or } g(b-x). \tag{3.17}$$

In particular, the constant 0 is excluded even when $F''(0) = 0$, because $F(0) > 0$ is inconsistent with the half-stress energy bound.

Write $\mathcal{O}(u)$ for the closure of the translates of u in $C^2_{\mathrm{loc}}(\mathbb{R}^2)$. The global $C^{2,\gamma}_b$ bound makes this family compact, and continuity of each fixed translation makes it translation invariant. Its elements inherit $-1 \le w \le 1$, $w_y \ge 0$, and (1.1), by the compactness and passage to the equation established in Section 2. For later use, an iterated translated limit is still a limit of translates of the original solution: every translate of $w \in \mathcal{O}(u)$ belongs to $\mathcal{O}(u)$, and closedness places each limit of those translates in $\mathcal{O}(u)$ as well.

For every $w \in \mathcal{O}(u)$, boundedness and vertical monotonicity define the pointwise limits

$$v^-(x) = \lim_{y \to -\infty} w(x,y), \qquad v^+(x) = \lim_{y \to +\infty} w(x,y). \tag{3.18}$$

Given any sequence $t_j \to +\infty$ or $t_j \to -\infty$, compactness gives a subsequential C^2_{loc} limit of $w(x, y+t_j)$. At every fixed (x,y) its value must be $v^+(x)$ or $v^-(x)$, respectively, by the defining monotone limit. Thus the limit is stationary and independent of the subsequence. Compactness consequently gives convergence of the entire family $w(\cdot, \cdot + t)$ in $C^2_{\mathrm{loc}}(\mathbb{R}^2)$ as $t \to \pm\infty$. The preceding closedness argument makes each end a stationary translated limit of the original u, so both ends belong to (3.17). In particular the traces converge locally on horizontal compact sets. If $w \not\equiv 1$, then $v^- \not\equiv 1$, since $v^-(x) \le w(x,y) \le 1$. Every other alternative in (3.17) has infimum -1; hence $\inf_x v^-(x) = -1$.

4. Minimum envelopes and uniform clearing

4.1. The one-dimensional balance identity.

Lemma 4.1 (Balance for monotone Lipschitz profiles)**.** *Let $0 < \alpha \le 1$, and let $a : \mathbb{R} \to [-1, 1]$ be Lipschitz and nondecreasing, with limits $a(-\infty) = -1$ and $a(+\infty) = 1$. Let $h = a'$ be its almost-everywhere derivative. Then $h \in L^1 \cap L^\infty \subset L^2$, and*

$$\int_{\mathbb{R}} (L_1 a) a' \, dy = 0. \tag{4.1}$$

For $\alpha < 1$, both the kernel integral defining $L_1 a$ and the integral in (4.1) are absolutely convergent. For $\alpha = 1$, $L_1 a = \mathcal{H} h \in L^2$ and (4.1) is the real L^2 pairing with the Hilbert transform, with the sign convention giving Fourier multiplier $|\xi|$ for $\mathcal{H}\partial_y$. Consequently no such a can satisfy

$$L_1 a - c a' + F'(a) \ge 0 \quad \text{almost everywhere}, \qquad c > 0. \tag{4.2}$$

Proof. One has $h \ge 0$ and $\int h = 2$. For $0 < \alpha < 1$ use

$$L_1 a(y) = C_{1,s} \int_0^\infty \frac{2a(y) - a(y+t) - a(y-t)}{t^{1+\alpha}} \, dt.$$

The bracket is bounded in absolute value by $C \min(t, 1)$. The resulting integral against the nonnegative measure $h(y)\, dy\, t^{-1-\alpha}\, dt$ is therefore absolutely convergent. For every fixed t, integration of the absolutely continuous product $a(y)a(y+t)$ gives

$$\int_{\mathbb{R}} \big[h(y) a(y+t) + a(y) h(y+t) \big] \, dy = 1 - 1 = 0.$$

After a change of variable in the second term, $\int h(y)[a(y+t) + a(y-t)]\, dy = 0$. Also $\int 2ah = [a^2]_{-\infty}^{+\infty} = 0$. Thus the inner integral of the symmetric second difference is zero. Fubini proves (4.1).

For $\alpha = 1$, define the Hilbert transform on $L^2(\mathbb{R})$ by $\widehat{\mathcal{H}h}(\xi) = -i \operatorname{sgn}(\xi) \widehat{h}(\xi)$. To identify the fractional operator distributionally, take $\varphi \in C_c^\infty(\mathbb{R})$, one has $\mathcal{H}\varphi = O(|y|^{-1})$ at infinity and $(\mathcal{H}\varphi)' = L_1 \varphi \in L^1$. Since a is bounded, integration by parts and L^2 skew-adjointness give

$$\langle L_1 a, \varphi \rangle = \int_{\mathbb{R}} a L_1 \varphi = - \int_{\mathbb{R}} h \mathcal{H} \varphi = \int_{\mathbb{R}} (\mathcal{H} h) \varphi.$$

Thus $L_1 a = \mathcal{H} h$ in distributions and belongs to L^2.

The same identity holds for the almost-everywhere principal value used below. With $C_{1,1/2} = 1/\pi$, integration by parts in the truncated singular integral gives

$$\begin{aligned} \frac{1}{\pi} \int_\varepsilon^\infty \frac{2a(y) - a(y+t) - a(y-t)}{t^2} \, dt = {} & \frac{2a(y) - a(y+\varepsilon) - a(y-\varepsilon)}{\pi \varepsilon} \\ & + \frac{1}{\pi} \int_\varepsilon^\infty \frac{h(y-t) - h(y+t)}{t} \, dt. \end{aligned}$$

The boundary term at infinity vanishes by boundedness of a. At its differentiability points the first term on the right tends to zero. The L^2 principal-value theorem for the Hilbert transform makes the second term converge almost everywhere to $\mathcal{H}h$. This proves the asserted almost-everywhere fractional principal value. Finally, skew-adjointness gives $\int_{\mathbb{R}} (L_1 a) h = \int_{\mathbb{R}} (\mathcal{H} h) h = 0$; the product is integrable by Cauchy–Schwarz.

Multiply (4.2) by $h \ge 0$ and integrate. Every term is integrable by the preceding argument and the chain rule, and

$$0 \le \int_{\mathbb{R}} (L_1 a - c a' + F'(a)) a' = -c \int_{\mathbb{R}} h^2 + F(1) - F(-1) = -c \int_{\mathbb{R}} h^2.$$

This forces $h = 0$, contradicting $\int h = 2$. □

4.2. **The horizontal minimum envelope.**

Lemma 4.2 (Horizontal infimum rigidity)**.** *For every $w \in \mathcal{O}(u)$, either $w \equiv 1$, or*

$$\inf_{x\in\mathbb{R}} w(x,y) = -1 \quad \textit{for every } y \in \mathbb{R}. \tag{4.3}$$

Proof. Assume $w \not\equiv 1$, and set $a(y) = \inf_x w(x,y)$. This is nondecreasing and globally Lipschitz, with Lipschitz constant at most $\|u_y\|_\infty$. Commuting the two infima and using the lower vertical end gives

$$a(-\infty) = \inf_y \inf_x w(x,y) = \inf_x v^-(x) = -1.$$

Suppose, for a contradiction, that $a \not\equiv -1$. Then $A := a(+\infty) > -1$. Choose $y_j \to +\infty$ and x_j with $w(x_j, y_j) \le a(y_j) + j^{-1}$. Passing to a subsequence of $w(x + x_j, y + y_j)$ gives a limit $W \in \mathcal{O}(u)$ satisfying $W \ge A$ everywhere and $W(0,0) = A$, because $a(y_j + t) \to A$ for each fixed t.

Monotonicity gives $W(0,t) = A$ for $t \le 0$. By the differentiated equation justified in Section 2, the nonnegative function $z = W_y$ solves

$$(L_2 - c\partial_y + F''(W))z = 0$$

classically. At $P = (0,-1)$ it vanishes, so $z_y(P) = 0$ and its equation gives

$$0 = L_2 z(P) = -C_{2,s} \int_{\mathbb{R}^2} \frac{z(Q)}{|P-Q|^{2+\alpha}}\, dQ.$$

The $C^{1,\gamma}$ bound and the zero minimum give $z(P+H) = O(|H|^{1+\gamma})$ with $1+\gamma > \alpha$; boundedness controls the tail. Thus the integral converges, and positivity of its kernel forces $z \equiv 0$. Thus W is stationary. Classification (3.17), together with $W \ge A > -1$, gives $W \equiv 1$ and $A = 1$. The envelope a therefore has the two limits $-1, 1$. We show that it satisfies (4.2). Fix a point y_0 where a is differentiable and its fractional integral exists. These conditions hold almost everywhere: when $\alpha < 1$, the Lipschitz bound makes that integral absolutely convergent everywhere; when $\alpha = 1$, Lemma 4.1 identifies it with an L^2 Hilbert transform almost everywhere.

Choose x_j with $w(x_j, y_0) \le a(y_0) + j^{-1}$ and pass to a limit V of the horizontal translates $w(x + x_j, t)$ in C^2_{loc}. Since the function $t \mapsto V(0,t) - a(t)$ has a differentiable minimum at y_0, one has

$$V(0,y_0) = a(y_0), \qquad V(x,t) \ge a(t) \quad \text{for all } (x,t), \qquad V_y(0,y_0) = a'(y_0).$$

Comparing the singular integrals, using the one-dimensional marginal of the normalized two-dimensional kernel, gives

$$L_2 V(0,y_0) \le L_1 a(y_0).$$

At $\alpha = 1$, compare first using the same circular cutoff $h^2 + t^2 > \varepsilon^2$ in the two-dimensional integrals of V and the lift $(x,y) \mapsto a(y)$. The difference integrand has one sign. For $|t| \ge \varepsilon$ the horizontal marginal is the full marginal. For $|t| < \varepsilon$, its additional part is

$$M_\varepsilon(t) = \int_{|h| > \sqrt{\varepsilon^2 - t^2}} (h^2 + t^2)^{-3/2}\, dh = \frac{2}{\varepsilon^2(1 + \sqrt{1 - t^2/\varepsilon^2})} \le \frac{2}{\varepsilon^2},$$

with its continuous value at $t = 0$. Pairing t and $-t$, the difference between this circular cutoff and the one-dimensional cutoff is bounded by

$$C\varepsilon^{-2} \int_0^\varepsilon |2a(y_0) - a(y_0 + t) - a(y_0 - t)|\, dt = o(1).$$

Thus the circular cutoff converges to $L_1 a(y_0)$ at the almost-everywhere points already selected. The classical integral of V also converges, so the comparison is legitimate. Substitution in the equation for V yields

$$L_1 a(y_0) - c a'(y_0) + F'(a(y_0)) \ge 0.$$

Lemma 4.1 now gives a contradiction. Hence $a \equiv -1$. □

Remark 4.3. The argument uses an infimum over the entire horizontal line, not a minimum over a finite interval. It is compatible with the tip of an upward two-arm wave: such a slice still approaches -1 at its remote horizontal ends. No upper envelope inequality is being substituted for the lower envelope inequality.

4.3. **Interior clearing and the no-merging principle.**

Lemma 4.4 (Uniform interior clearing)**.** *For every $-1 < a \le b < 1$ there exists $R(a,b) \ge 1$ such that, for every $w \in \mathcal{O}(u)$ and $x_0, y \in \mathbb{R}$,*

$$w(x,y) \ge a \quad (|x - x_0| \le R(a,b)) \quad \Longrightarrow \quad w(x_0,y) > b. \tag{4.4}$$

In particular, every interval on which $a \le w(\cdot,y) \le b$ has length at most $2R(a,b)$.

Proof. Otherwise choose violating profiles in $\mathcal{O}(u)$ and translate their intervals, with radii tending to infinity, to $(0,0)$. Compactness and translation invariance give a limit $w \in \mathcal{O}(u)$ with $w(x,0) \ge a$ for every x and $w(0,0) \le b < 1$. The latter inequality rules out $w \equiv 1$, while the former contradicts (4.3) since $a > -1$. An interval of length greater than $2R(a,b)$ contains a closed subinterval of radius $R(a,b)$, so the interval assertion follows from (4.4). □

Lemma 4.5 (Propagation of proximity to a well on fixed balls)**.** *Given $D > 0$ and $b < 1$, there exists $\eta = \eta(D,b) \in (-1,1)$ such that, for every $w \in \mathcal{O}(u)$ and $P \in \mathbb{R}^2$,*

$$w(P) \ge \eta \quad \Longrightarrow \quad w > b \quad \text{on } \overline{B_D(P)}.$$

Proof. Otherwise there are $w_j \in \mathcal{O}(u)$ and points P_j, Q_j with $w_j(P_j) \to 1$, $|Q_j - P_j| \le D$, and $w_j(Q_j) \le b$. Compactness of the translated orbit and of the closed ball gives, after a subsequence,

$$w_j(P_j + \cdot) \longrightarrow v \quad \text{in } C^2_{\mathrm{loc}}, \qquad Q_j - P_j \longrightarrow Q.$$

Thus $v(0) = 1$ and $v(Q) \le b$. The nonnegative function $z = 1 - v$ vanishes at the origin and satisfies

$$(L_2 - c\partial_y + B)z = 0, \qquad B(X) = \int_0^1 F''(1 + t(v(X) - 1))\, dt.$$

The coefficient is bounded. The zero-minimum argument in the proof of Lemma 4.2 gives $z \equiv 0$, contradicting $v(Q) \le b < 1$. □

Fix $R_0 \ge 1$ for Lemma 4.4 with $a = 0, b = 1/2$. Apply Lemma 4.5 with $D = 2R_0$ and $b = 1/2$, increasing its threshold if necessary to fix $\eta \in (1/2, 1)$ such that

$$u(p,y) \ge \eta \quad \Longrightarrow \quad u(x,y) > 1/2 \quad (|x - p| \le 2R_0). \tag{4.5}$$

Proposition 4.6 (Two high-phase points cannot enclose a nonpositive point)**.** *For the fixed η above, if $p < q$ and $u(p,y_0), u(q,y_0) \ge \eta$, then*

$$u(x,y_0) > 0 \quad (p \le x \le q). \tag{4.6}$$

Consequently at every height all points with $u \ge \eta$ belong to one connected component of $\{u(\cdot,y) > 0\}$, whenever that high-phase set is nonempty.

Proof. Suppose $m(y_0) := \min_{x \in [p,q]} u(x,y_0) \le 0$. The function m is Lipschitz with constant at most $\|u_y\|_\infty$. By monotonicity and Dini's theorem, $u(\cdot,y) \to 1$ uniformly on $[p,q]$ as $y \to +\infty$. Hence $m(y_*) = 0$ for some $y_* \ge y_0$.

Let x_* be a minimizer at this height. The endpoint values remain at least η by vertical monotonicity. Equation (4.5) therefore excludes a zero within distance $2R_0$ of either endpoint, so $[x_* - R_0, x_* + R_0] \subset [p,q]$. On this interval $u(\cdot,y_*) \ge 0$, whereas $u(x_*,y_*) = 0$. Lemma 4.4 with $a = 0$ and $b = 1/2$ gives a contradiction. □

Remark 4.7. The conclusion is a coarse convexity statement for two fixed phase levels. It does not assert that every superlevel set is an interval or that u_x already has a prescribed sign. Small low-amplitude positive components have not yet been excluded.

5. Uniform two-arm geometry and transverse localization

5.1. **Rigidity of bounded-width single fronts.** The next theorem allows its interface center to be unbounded. This is important when it is applied to escaping translated limits: a trap between two fixed translates of a stationary layer has not been proved.

Lemma 5.1 (Bounded negative-set comparison)**.** *Let $0 < \theta < 1$ and let $z \in C_b^{1,\theta}(\mathbb{R}^2)$ be real valued and satisfy*

$$(L_2 - c\partial_y + B)z = 0 \quad \text{in } \mathbb{R}^2,$$

where B is bounded and uniformly continuous. If $B \geq \mu > 0$ at every point where $z < 0$, then $z \geq 0$.

Proof. Put $z_- = (|z| - z)/2$. Kato's inequality (2.5) and the equation give

$$(L_2 - c\partial_y + \mu)z_- \leq (\mu - B)z_- \leq 0 \quad \text{in distributions.}$$

The last inequality uses $B \geq \mu$ on $\{z < 0\}$ and $z_- = 0$ elsewhere. Since z_- is bounded, Lemma 2.1 gives $z_- \leq 0$. Thus $z_- = 0$ and $z \geq 0$. □

Theorem 5.2 (Coarsely Lipschitz single-front rigidity)**.** *Let $0 < \alpha \leq 1$. Let w be a bounded uniformly $C^{2,\gamma}$ solution of (1.1), with $-1 < w < 1$ and $w_y \geq 0$. Suppose there is a function $a : \mathbb{R} \to \mathbb{R}$ such that*

$$|a(t) - a(y)| \leq K(1 + |t - y|) \tag{5.1}$$

and, for every $\delta > 0$, a finite L_δ with

$$\begin{cases} w(x,y) < -1 + \delta, & x < a(y) - L_\delta, \\ w(x,y) > 1 - \delta, & x > a(y) + L_\delta. \end{cases} \tag{5.2}$$

Then $w(x,y) = g(x - b)$ for some b. The reversed phase orientation gives $w(x,y) = g(b - x)$. No boundedness of a is assumed.

Proof. We first record how the two phases persist in the translated limits used below. Suppose $|x_j - a(y_j)| \leq C_0$ and $w_j(x,t) = w(x + x_j, t + y_j) \to W(x,t)$ in C^2_{loc}. For every $T > 0$, (5.1) gives

$$|a(y_j + t) - x_j| \leq C_0 + K(1 + T) =: D_T \qquad (|t| \leq T).$$

Apply (5.2) with $\delta/2$ before taking the limit. It follows that

$$\begin{cases} W(x,t) < -1 + \delta, & x < -D_T - L_{\delta/2}, \\ W(x,t) > 1 - \delta, & x > D_T + L_{\delta/2}, \end{cases} \qquad |t| \leq T.$$

Thus W retains both horizontal end states, uniformly on bounded temporal intervals. It also inherits the equation, the global regularity bounds, and $W_y \geq 0$ by Section 2.

Horizontal sliding. Choose $0 < \delta_0 < 1$ and $\mu > 0$ such that $F'' \geq \mu$ on $[-1, -1 + \delta_0] \cup [1 - \delta_0, 1]$. Set $L = L_{\delta_0}$ and $z_h(x,y) = w(x + h, y) - w(x,y)$ for $h > 0$. If $h > 2L$ and $z_h < 0$, both compared values lie in the same endpoint interval. In fact, if $x < a(y) - L$, the unshifted value is near -1 and the shifted value is still smaller. If $x \geq a(y) - L$, then $x + h > a(y) + L$, so the shifted value is near 1 and the unshifted value is still larger. The divided-difference coefficient in the equation for z_h is therefore at least μ at negative points. Lemma 5.1 gives $z_h \geq 0$ for all sufficiently large h.

Put

$$\mathcal{S} = \{h > 0 : z_t \geq 0 \text{ everywhere for every } t \geq h\}, \qquad h_* = \inf \mathcal{S}.$$

This set is nonempty and upward closed. Consequently $z_t \geq 0$ for $t > h_*$, and continuity in the shift gives $z_{h_*} \geq 0$ when $h_* > 0$. Suppose $h_* > 0$. For all sufficiently large j, $h_* - j^{-1} > 0$ is not in $\mathcal{S}$, so there is $h_j \in [h_* - j^{-1}, h_*)$ for which z_{h_j} is negative somewhere. There must be a negative point where the two compared values are not in the same endpoint interval; otherwise Lemma 5.1 would rule out all negative values. At such a point $P_j = (x_j, y_j)$, $|x_j - a(y_j)| \leq L + \sup_j h_j$; outside this range both points are in the same endpoint interval. Also

$$0 \leq z_{h_*}(P_j) \leq \|w_x\|_\infty |h_* - h_j| \longrightarrow 0.$$

Translate P_j to the origin. The phase-retention observation gives a limit W with both horizontal end states. The nonnegative difference $W(x+h_*,y)-W(x,y)$ solves a linear equation with bounded reaction coefficient and has a zero at the origin. The zero-minimum argument from Lemma 4.2 makes this difference vanish identically. This makes W periodic in x, contradicting its different limits at $x=\pm\infty$ on the slice $y=0$. Therefore $h_*=0$ and $w_x\geq 0$. The differentiated equation from Section 2 and the same zero-minimum argument give $w_x>0$: otherwise w_x would vanish identically, contradicting the two horizontal end states.

A uniform positive derivative in the transition core. For every fixed $0<\delta<1$ there is $m_\delta>0$ such that

$$w_x(x,y)\geq m_\delta \quad \text{if } |w(x,y)|\leq 1-\delta. \tag{5.3}$$

Otherwise translate core points where $w_x\to 0$. They stay a bounded distance from $a(y)$ by (5.2). The same compactness observation gives a limit with both horizontal end states and a nonnegative horizontal derivative vanishing at the origin. Its linearized equation and the zero-minimum argument give a contradiction. The derivative equation is classical by Section 2, including when $\alpha=1$.

The maximal admissible slope. Let $\mathcal{C}=\{|w|\leq 1-\delta_0\}$; each horizontal slice meets this set by the two end states. By (5.3),

$$p_*:=\sup_{\mathcal{C}}\frac{w_y}{w_x}<\infty.$$

The function $z=p_*w_x-w_y$ is nonnegative on $\mathcal{C}$ and solves $(L_2-c\partial_y+F''(w))z=0$. If it is negative, the point is outside $\mathcal{C}$, where $F''(w)\geq\mu$. Lemma 5.1 gives $z\geq 0$ everywhere. If $p_*>0$, choose $P_j\in\mathcal{C}$ with $w_y(P_j)/w_x(P_j)\to p_*$. Since w_x is bounded,

$$z(P_j)=w_x(P_j)\left(p_*-\frac{w_y(P_j)}{w_x(P_j)}\right)\longrightarrow 0.$$

The core points stay within bounded distance of $a(y_j)$, so their translated limit W retains both horizontal end states. The function $p_*W_x-W_y$ is nonnegative, solves the linearized equation, and vanishes at the origin. The zero-minimum argument yields

$$W_y=p_*W_x \quad \text{everywhere}, \qquad W(x,y)=H(x+p_*y).$$

The profile H is bounded, Lipschitz, and nondecreasing, with $H(\pm\infty)=\pm 1$. Rotation invariance, scaling, and kernel marginalization reduce the equation to

$$(1+p_*^2)^{\alpha/2}L_1H-cp_*H'+F'(H)=0.$$

Multiplication by H' is legitimate by Lemma 4.1. Its balance identity and $F(1)=F(-1)$ give $0=-cp_*\int(H')^2<0$, since $p_*>0$ and $\int H'=2$. Thus $p_*=0$, so $w_y=0$. The stationary monotone layer is $g(x-b)$ by uniqueness. Horizontal reflection proves the other orientation without reflecting values of w or imposing evenness on F. □

5.2. **Lateral expansion and exterior clearing.** Let η be fixed as in Proposition 4.6, and set

$$\eta_1=\frac{1+\eta}{2}, \qquad B=\|u_y\|_\infty>0, \qquad \tau=\frac{\eta_1-\eta}{2B}, \qquad R_1\geq R(0,\eta_1). \tag{5.4}$$

If $u(x,y+\tau)>\eta_1$, then $u(x,y)>\eta$.

Whenever the high-phase set is nonempty, let $I(y)$ be the unique positive component containing it, as supplied by Proposition 4.6. If $I(y)$ exists, then it exists at every later height and $I(y)\subset I(y+t)$ for $t\geq 0$: a high-phase point persists, and the earlier interval remains connected and positive by vertical monotonicity.

For an open interval $J=(a,b)$, define $J^{-R}=(a+R,b-R)$, with infinite endpoints left unchanged and the result interpreted as empty when $a+R\geq b-R$. If $x\in I(y+\tau)^{-R_1}$, then $[x-R_1,x+R_1]\subset I(y+\tau)$. Interior clearing and (5.4) give

$$u(x,y+\tau)>\eta_1, \qquad u(x,y)>\eta_1-B\tau=\frac{\eta_1+\eta}{2}>\eta.$$

Consequently, whenever $I(y)$ is defined,

$$I(y+\tau)^{-R_1} \subset I(y). \tag{5.5}$$

Assume now that $I(y)$ has length greater than $2R_1$. By nesting, all later erosions are nonempty. If $I(y) = (l(y), r(y))$ is finite, (5.5) forces $I(y+\tau)$ to be finite as well. Comparing endpoints and using nesting gives

$$l(y) - R_1 \le l(y+\tau) \le l(y), \qquad r(y) \le r(y+\tau) \le r(y) + R_1. \tag{5.6}$$

The same argument applies to the finite endpoint of a half-line. Under this same length assumption, for arbitrary $t > 0$ iterate with $n = \lceil t/\tau \rceil$ and use $I(y+t) \subset I(y+n\tau)$. It follows that

$$I(y+t) \subset I(y) + [-C(1+t), C(1+t)], \qquad C = R_1 \max\{1, \tau^{-1}\}.$$

The inclusion is also immediate at $t = 0$.

Lemma 5.3 (Uniform exterior clearing from controlled expansion)**.** *Suppose a family $I(y)$ as above is defined on $J = \mathbb{R}$ or $J = [Y, \infty)$, contains every point with $u \ge \eta$, and satisfies the expansion bound*

$$I(y+t) \subset I(y) + [-C(1+t), C(1+t)] \qquad (t \ge 0,\ y \in J). \tag{5.7}$$

For every $\delta > 0$ there is $A_\delta < \infty$ such that

$$y \in J, \quad \operatorname{dist}(x, I(y)) \ge A_\delta \quad \Longrightarrow \quad u(x,y) < -1 + \delta. \tag{5.8}$$

Proof. Suppose instead that there are $y_j \in J$ and x_j with distances to $I(y_j)$ tending to infinity and $u(x_j, y_j) \ge -1 + \delta$. The domain J is forward invariant. For every fixed $M, T > 0$, (5.7) gives, uniformly for $|X| \le M$ and $0 \le t \le T$,

$$\operatorname{dist}(x_j + X, I(y_j + t)) \ge \operatorname{dist}(x_j, I(y_j)) - M - C(1+T) \longrightarrow \infty.$$

Outside $I(y_j + t)$ the solution is below η, since this component contains every high-phase point. Thus a translated limit $w \in \mathcal{O}(u)$ satisfies $w(x,t) \le \eta$ for every x and $t \ge 0$, while $w(0,0) \ge -1 + \delta$. Its upper stationary vertical end is at most $\eta < 1$ everywhere. Among (3.17), only the constant -1 has that property. Hence $v^+ \equiv -1$. Monotonicity gives $w \le v^+ = -1$, contradicting its value at the origin. □

Proposition 5.4 (There are no unbounded positive components)**.** *For every finite y, every connected component of $\{x : u(x,y) > 0\}$ is a bounded interval.*

Proof. An all-positive slice contradicts Lemma 4.2. Suppose a component at some y_0 is a right half-line. Interior clearing gives $u(x, y_0) \to 1$ as $x \to +\infty$. For every $D > 0$ and $\delta > 0$, Lemma 4.5 transfers sufficient proximity to 1 at (x, y_0) to its closed radius-D ball. Consequently

$$u(x,y) \longrightarrow 1 \quad (x \to +\infty)$$

uniformly for $|y - y_0| \le D$, including heights below y_0. At every height there is therefore an unbounded right positive component. Proposition 4.6 makes it the unique component containing all high-phase points. Write it as $I(y) = (a(y), \infty)$, where $a(y)$ is finite since an all-positive slice is impossible.

These half-lines are nested. Formula (5.5) bounds the motion of their finite endpoint on every interval of length τ, and thus a satisfies (5.1). Lemma 5.3 applies on all of $\mathbb{R}$ and gives the uniform negative phase to the left. Interior clearing gives the uniform positive phase to the right. Thus u is a bounded-width single front satisfying Theorem 5.2. It would be stationary, contrary to $u_y > 0$. The left-half-line case follows by horizontal reflection. □

5.3. **Uniform phase widths and properness.**

Proposition 5.5 (A bounded expanding positive component)**.** *At sufficiently large heights the positive component $I(y)$ containing the origin is a bounded interval, is nested as y increases, and contains all points where $u \geq \eta$. Its endpoints satisfy the coarse expansion bound (5.6), tend to the opposite horizontal infinities, and obey the uniform interior and exterior phase bounds (5.9) below. Moreover $u(x,y) \to -1$ as $|x| \to \infty$ uniformly on each region $y \leq T$, and (1.8) holds.*

Proof. Choose Y_0 so large that

$$u(x, Y_0) > \eta_1 \quad \text{for } |x| \leq 2R_1 + 1.$$

This uses only the pointwise positive vertical limit and its uniformity on a fixed compact interval. For every $y \geq Y_0$, let $I(y)$ be the positive component containing that interval. It contains all high-phase points by Proposition 4.6; it is finite by Proposition 5.4. Write

$$I(y) = (l(y), r(y)).$$

Its erosion never becomes empty, so (5.6) and (5.7) hold on $[Y_0, \infty)$. Lemma 5.3 and interior clearing with $a = 0$, $b = 1 - \delta$ imply that, for $0 < \delta < 1$, some $K_\delta < \infty$ satisfies

$$\begin{cases} u(x,y) < -1 + \delta, & x < l(y) - K_\delta \text{ or } x > r(y) + K_\delta, \\ u(x,y) > 1 - \delta, & l(y) + K_\delta < x < r(y) - K_\delta, \end{cases} \qquad y \geq Y_0. \tag{5.9}$$

For larger δ, use the bound for any smaller parameter in $(0,1)$. Moreover,

$$l(y) \to -\infty, \qquad r(y) \to +\infty \qquad (y \to +\infty), \tag{5.10}$$

because every fixed compact interval eventually lies in the positive component containing the origin.

For each fixed T, apply (5.9) on the slice $T_0 = \max\{T, Y_0\}$. It gives $u(x, T_0) \to -1$ as $|x| \to \infty$. Monotonicity and $u \geq -1$ then give this lateral convergence uniformly over all $y \leq T$. For a given $\delta > 0$, discard the horizontal tails using the slice $T = Y_0$. On the remaining compact horizontal interval use the negative vertical limit and Dini's theorem. This proves the uniform negative limit in (1.8). For every x, strict vertical monotonicity and the two vertical limits give a unique zero $y = \gamma(x)$. The implicit function theorem gives $\gamma \in C^2$. At any fixed height T, the lateral convergence just proved makes $u(x,T) < 0$ for all sufficiently large $|x|$. Hence $\gamma(x) > T$, proving $\gamma(x) \to +\infty$ and completing (1.8).

□

5.4. **Stationarity of every escaping transition.**

Proof of Theorem 1.2. Consider (x_j, y_j) with $|x_j| \to \infty$ and $|u(x_j, y_j)| \leq 1 - \delta$ for some fixed $\delta > 0$. Lateral convergence uniform below every fixed height gives $y_j \to \infty$. The phase bounds (5.9) place x_j within a fixed distance of $l(y_j)$ or $r(y_j)$. After a subsequence, suppose $x_j \to +\infty$. Since $l(y_j) \to -\infty$, it must then be the right endpoint: $|x_j - r(y_j)| \leq C_\delta$. The left-escaping case follows by horizontal reflection.

Pass to a C^2_{loc} limit w of the translates $u(x_j + X, y_j + t)$, and put $a_j(t) = r(y_j + t) - x_j$ wherever $y_j + t \geq Y_0$. These domains exhaust $\mathbb{R}$. The centers are nondecreasing, bounded at $t = 0$, and satisfy a common coarse bound (5.1); hence they are bounded on every compact time interval. A diagonal application of Helly's theorem gives convergence at every continuity point of a monotone limiting function. Its discontinuity set is countable. Refining the subsequence at those points gives limits $a_j(t) \to a(t)$ for every $t \in \mathbb{R}$. The values selected at the discontinuities lie between the corresponding one-sided limits, so a remains monotone; the common coarse bound passes to it pointwise. Also $l(y_j + t) - x_j \to -\infty$ for every fixed t.

For each $\varepsilon > 0$ use (5.9) with $\varepsilon/2$, and put $L_\varepsilon = K_{\varepsilon/2} + 1$. At any fixed time t, passage to the limit gives

$$w(X,t) > 1 - \varepsilon \quad (X < a(t) - L_\varepsilon), \qquad w(X,t) < -1 + \varepsilon \quad (X > a(t) + L_\varepsilon).$$

The widths are independent of t. In the first inequality the opposite endpoint has escaped to $-\infty$; the use of $\varepsilon/2$ makes both limiting inequalities strict. The value $w(0,0)$ remains separated from the two wells. If w attained either well elsewhere, the equation at that global extremum would force it to be the corresponding constant, a contradiction. Thus $-1 < w < 1$, and $w_y \geq 0$ with the uniform regularity inherited from the translated orbit. Theorem 5.2 therefore gives $w(X,t) = g(b-X)$. A left-escaping limit is $g(X-b)$. For zero-centered translates the value at the origin forces $b = 0$. Every subsequence has the same possible normalized limit, so orbit compactness upgrades this conclusion to convergence of the full zero-centered sequences in (1.9).

If (1.10) failed, there would be $|x_j| \to \infty$ and y_j with $u_y(x_j, y_j) \geq \varepsilon > 0$. Set $M = \|u_{yy}\|_\infty$ and $h = \varepsilon/(2(1+M))$. On $|t - y_j| \leq h$ the derivative is at least $\varepsilon/2$. Integrating in both vertical directions and using $|u| \leq 1$ gives

$$-1 + \frac{\varepsilon h}{2} \leq u(x_j, y_j) \leq 1 - \frac{\varepsilon h}{2}.$$

Thus this is a transition sequence, whose stationary limit contradicts the derivative lower bound. Hence (1.10) holds.

At every sufficiently remote right zero, u_x is uniformly negative, and at every sufficiently remote left zero it is uniformly positive, by (1.9). At high heights there are no zeros in a fixed compact horizontal interval, since $u \to 1$ there uniformly. Thus every right zero is a strictly decreasing crossing and every left zero a strictly increasing crossing. Two decreasing crossings on the right would require a return from negative to positive values between them, and hence a zero with nonnegative derivative. This is impossible; the reflected argument applies on the left. The finite positive component containing the origin already has a zero endpoint on each side. These are therefore the only zeros, with $\ell(y) = l(y)$ and the same right endpoint $r(y)$. The implicit function theorem now yields C^2 functions ℓ, r, with

$$\ell' = -\frac{u_y}{u_x}(\ell(y), y) < 0, \qquad r' = -\frac{u_y}{u_x}(r(y), y) > 0.$$

The denominator stays bounded away from zero and (1.10) makes both velocities tend to zero. This proves (1.11).

For completeness, (1.12) follows by a three-region argument. On fixed-width neighborhoods of either zero, the preceding layer compactness gives convergence to the appropriate normalized copy of g. Away from those neighborhoods, (5.9) gives uniform proximity to the appropriate wells, as do the tails of the comparison layers. First choose the neighborhood width large and then send $y \to \infty$. This proves uniform convergence of the functions. To obtain convergence of horizontal derivatives, write

$$E_y(x) = u(x,y) - g(x - \ell(y)) - g(r(y) - x) + 1.$$

The global regularity bounds for u and g give $\sup_y \|E_y''\|_\infty \leq M_2 < \infty$. Taylor's formula, at any horizontal increment $h > 0$, yields

$$\|E_y'\|_\infty \leq \frac{2\|E_y\|_\infty}{h} + \frac{M_2 h}{2}.$$

If $M_2 > 0$, optimize in h to obtain $\|E_y'\|_\infty^2 \leq 4M_2\|E_y\|_\infty \to 0$; if $M_2 = 0$, boundedness of E_y gives $E_y' = 0$. This proves (1.12) and completes the geometry theorem.

□

5.5. **An integrable transverse derivative envelope.** The geometry also yields a useful tail estimate before normalization.

Proposition 5.6 (First derivatives localized to two moving tubes)**.** *After increasing Y in Theorem 1.2, extend ℓ, r constantly to $y < Y$. Then, for some $C < \infty$,*

$$|u_x(x,y)| + u_y(x,y) \leq C\left[\langle x - \ell(y)\rangle^{-1-\alpha} + \langle x - r(y)\rangle^{-1-\alpha}\right] \quad \text{for all } (x,y) \in \mathbb{R}^2. \tag{5.11}$$

In particular,

$$\sup_y \int_{\mathbb{R}} (|u_x| + u_y)(x,y)\,dx < \infty, \tag{5.12}$$

and the horizontal derivative mass outside the union of the two radius-R core intervals is $O(R^{-\alpha})$, uniformly in y.

Proof. The extended centers are globally Lipschitz, since their derivatives are uniformly small above a sufficiently large Y and they are constant below it. Choose a well neighborhood on which $F'' \geq 2\mu > 0$. By (5.9), the complement of these two well neighborhoods lies, for $y \geq Y$, within fixed-width tubes about $\ell(y), r(y)$. For $y \leq Y$, lateral convergence uniform below Y puts the same transition set in a fixed compact horizontal interval. Increasing the tube widths covers this interval with the two extended constant tubes. Thus the entire transition set lies in

$$\mathcal{T}_1 \cup \mathcal{T}_2, \qquad \mathcal{T}_i = \{(x,y) : |x - a_i(y)| \leq W\}, \quad a_1 = \ell,\ a_2 = r.$$

For each derivative $v = \partial_i u$, the differentiated equation and Kato's inequality give

$$(L_2 - c\partial_y + \mu)|v| \leq C(\mathbf{1}_{\mathcal{T}_1} + \mathbf{1}_{\mathcal{T}_2})$$

in distributions. Here the global derivative bound controls the source on the transition set, and the coefficient outside it has the required positive sign.

The positive massive resolvent has kernel

$$G_c(Z) = \int_0^\infty e^{-\mu t} p_t^{(2)}(Z + cte_y)\,dt, \qquad \int_{\mathbb{R}^2} G_c = \mu^{-1}.$$

Its large-distance bound is

$$G_c(Z) \leq C|Z|^{-2-\alpha} \qquad (|Z| \geq 1). \tag{5.13}$$

Indeed, split the time integral at $|Z|/(2(1+c))$. Below that point, $|Z + cte_y| \geq |Z|/2$ and the stable-kernel bound $p_t^{(2)}(H) \leq Ct/(t^{1/\alpha} + |H|)^{2+\alpha}$ gives the claimed power after integration against $e^{-\mu t}$. Above it, use $p_t^{(2)} \leq Ct^{-2/\alpha}$ and the exponential killing; for large $|Z|$ this contribution is exponentially small. Increasing C handles the remaining bounded range $|Z| \geq 1$. The function $\Psi = CG_c * (\mathbf{1}_{\mathcal{T}_1} + \mathbf{1}_{\mathcal{T}_2})$ is bounded and continuous, since $G_c \in L^1$, and solves the corresponding massive source equation by (2.4). Thus $(L_2 - c\partial_y + \mu)(|v| - \Psi) \leq 0$ in distributions, and Lemma 2.1 gives $|v| \leq \Psi$.

To estimate one tube, let $\rho = |x - a_i(y)|$ and let K be a Lipschitz constant for a_i. For $(x', t) \in \mathcal{T}_i$,

$$\rho \leq |x - x'| + W + K|y - t|.$$

Write $d = |(x,y) - (x',t)|$. Cauchy–Schwarz and the Lipschitz bound give

$$d \geq \frac{\rho - W}{\sqrt{1 + K^2}}, \qquad d \geq |y - t|.$$

For $\rho \geq 2W$, these imply $d \geq c_K(\rho + |y - t|)$. Increasing the lower threshold for ρ also ensures $d \geq 1$, so (5.13) applies throughout the tube integration. Integration of (5.13) over the tube, whose horizontal cross-section has length $2W$, yields

$$(G_c * \mathbf{1}_{\mathcal{T}_i})(x,y) \leq C \int_{\mathbb{R}} (\rho + |y - t|)^{-2-\alpha}\,dt \leq C\rho^{-1-\alpha}.$$

For bounded ρ, the total mass of G_c suffices. Summing the two tube bounds proves (5.11). The mass assertions follow by integrating each envelope: $\int_{\mathbb{R}} \langle t \rangle^{-1-\alpha}\,dt < \infty$ and $\int_{|t|>R} \langle t \rangle^{-1-\alpha}\,dt \leq CR^{-\alpha}$ for $R \geq 1$. Outside the union of the two radius-R cores, both distances satisfy this last restriction. □

Remark 5.7 (Absolute versus normalized tails). The envelope (5.11) concerns the original solution. Dividing an error by an amplitude tending to zero can destroy its uniform tail bound. In particular, it is not legitimate to apply a Liouville theorem to an arbitrary normalized local limit using this estimate alone. The proof in Section 7 instead localizes the exact reflection equation and proves tightness at the normalization scale.

6. A homogeneous linearized Liouville theorem

The argument in this section is linear and independent of the two-arm construction. Its growth hypothesis is accompanied by an explicit fractional tail condition. At low order, a bound by $C(1+|y|)$ alone does not ensure that a two-dimensional function belongs to $\mathcal{L}_\alpha$.

6.1. **A localized temporal moment.**

Proposition 6.1 (A compact transverse source has a finite temporal moment)**.** *Let $0<\alpha<2$, $\mu>0$, $c\in\mathbb{R}$, and $0\le\eta\in C_c^\infty(\mathbb{R})$. Put $S(x,y)=\eta(x)(1+|y|)$ and*

$$Z(X)=\int_0^\infty e^{-\mu\tau}(P_\tau S)(X+c\tau e_y)\,d\tau. \tag{6.1}$$

Then Z is finite, belongs to $\mathcal{L}_\alpha(\mathbb{R}^2)$, solves $(L_2-c\partial_y+\mu)Z=S$ in distributions, and satisfies

$$Z(x,y)\le C\left((1+|y|)\langle x\rangle^{-1-\alpha}+\langle x\rangle^{-\alpha}\right). \tag{6.2}$$

Consequently, for every finite $p>1/\alpha$ with $p\ge1$,

$$\|Z(\cdot,y)\|_{L^p_x}\le C_p(1+|y|). \tag{6.3}$$

If $h\ge0$ belongs to $\mathcal{L}_\alpha$ and $(L_2-c\partial_y+\mu)h\le CS$, then $h\le CZ$ almost everywhere, and pointwise if h is continuous.

Proof. In the convolution formula, bound $1+|y+c\tau-v|\le1+|y|+|c|\tau+|v|$. The first term gives $(1+|y|)B(x)$, where marginalization gives

$$B(x)=\int_0^\infty e^{-\mu\tau}(p_\tau^{(1)}*\eta)(x)\,d\tau.$$

The mass bound controls B on bounded sets, and (2.2) gives $B(x)\le C\langle x\rangle^{-1-\alpha}$ outside the support of η. The term with $|c|\tau$ satisfies the same bound, with a different constant: its time integral has the additional integrable factor τ.

For the remaining term, (2.2) in two dimensions gives

$$\int_{\mathbb{R}}|v|p_\tau^{(2)}(r,v)\,dv\le C\tau(\tau^{1/\alpha}+|r|)^{-\alpha}. \tag{6.4}$$

Indeed $\tau^{1/\alpha}+\sqrt{r^2+v^2}$ is comparable to $\tau^{1/\alpha}+|r|+|v|$, and the integral of $v(A+v)^{-2-\alpha}$ over $(0,\infty)$ is a constant times $A^{-\alpha}$. After convolution with η, the right side is uniformly bounded for x in a fixed compact set, because $\tau(\tau^{1/\alpha}+|r|)^{-\alpha}\le1$. For $|x|$ larger than twice the source radius, it is at most $C\tau|x|^{-\alpha}$. Integration against $e^{-\mu\tau}$ proves (6.2). Importantly, the horizontal source is convolved before any singular pointwise resolvent estimate is used.

The bound (6.2) gives the tail-class membership directly. For example,

$$\int_{\mathbb{R}}\frac{|y|}{(1+x^2+y^2)^{1+\alpha/2}}\,dy=C_\alpha\langle x\rangle^{-\alpha},$$

so the first term has an integrable x majorant $C\langle x\rangle^{-1-2\alpha}$. The identity

$$\int_{\mathbb{R}}(1+x^2+y^2)^{-1-\alpha/2}\,dy=C_\alpha\langle x\rangle^{-1-\alpha}$$

gives integrable majorants for the two remaining terms as well. For clarity, put $U_\tau(X)=e^{-\mu\tau}(P_\tau S)(X+c\tau e_y)$ and $Z_{\varepsilon,T}=\int_\varepsilon^T U_\tau\,d\tau$. The source belongs to $\mathcal{L}_\alpha$, since its horizontal support is compact and its vertical growth is linear. The approximate-identity argument in Lemma 2.1, applied with a compact smooth test φ replaced by $\varphi(\,\cdot-c\varepsilon e_y)$, gives $U_\varepsilon\to S$ in distributions as $\varepsilon\downarrow0$. On the other hand, the marginal mass bound and (6.4) give

$$U_\tau(x,y)\le Ce^{-\mu\tau}(1+|y|+\tau),$$

so $U_T\to0$ locally uniformly as $T\to\infty$. For each $\tau>0$, the weighted kernel bounds make U_τ continuous. The displayed bound is integrable in τ, locally uniformly in (x,y), so dominated

convergence gives continuity of Z. For $0 < \varepsilon < T < \infty$, semigroup differentiation against compact tests is legitimate by the weighted kernel bounds and yields

$$(L_2 - c\partial_y + \mu)Z_{\varepsilon,T} = U_\varepsilon - U_T.$$

Moreover $0 \le Z_{\varepsilon,T} \le Z$, and the already proved membership $Z \in \mathcal{L}_\alpha$ gives $Z_{\varepsilon,T} \to Z$ in that weighted L^1 space. The adjoint operator sends every compact smooth test to a function bounded by $C\langle X\rangle^{-2-\alpha}$, so this convergence permits passage on the left side. Passing to both endpoints proves the stated resolvent equation in distributions. The L^p claim is immediate from $\alpha p > 1$. Finally apply Lemma 2.1 to $h - CZ$. □

The decay term $\langle x\rangle^{-\alpha}$ need not be in L^2 when $\alpha \le 1/2$. The argument below uses a finite $p > \max\{2, 1/\alpha\}$ instead. No unrestricted first moment of the fractional heat kernel is invoked.

6.2. **Invertibility at nonzero temporal frequency.**

Lemma 6.2 (Nonzero-frequency invertibility on a finite L^p space)**.** *Let $0 < \alpha \le 1$, $2 < p < \infty$, $c \ne 0$, and let V be real and bounded, with $V \ge 2\mu > 0$ outside a compact interval. For every real $\zeta \ne 0$,*

$$A_\zeta = (-\partial_{xx} + \zeta^2)^{\alpha/2} + V - ic\zeta : H^{\alpha,p}(\mathbb{R}) \longrightarrow L^p(\mathbb{R})$$

is invertible. Its inverse depends smoothly on ζ on compact sets avoiding zero. Here $H^{\alpha,p}$ is the Bessel-potential space.

Proof. Set $B_\zeta = (-\partial_{xx}+\zeta^2)^{\alpha/2}$. Gaussian subordination gives a positive convolution semigroup with mass $e^{-t|\zeta|^\alpha}$. In particular its L^p norm is at most one. Choose $\chi \in C_c^\infty(\mathbb{R})$, $\chi \ge 0$, so that $V_0 = V + \chi \ge \mu$. The product formula for bounded multiplication perturbations gives

$$\|e^{-t(B_\zeta+V_0)}\|_{L^p\to L^p} \le e^{-\mu t}.$$

Thus

$$T_\zeta^{-1} f = \int_0^\infty e^{ic\zeta t} e^{-t(B_\zeta+V_0)} f \, dt, \qquad T_\zeta = B_\zeta + V_0 - ic\zeta,$$

is a bounded inverse on L^p. Its domain is $H^{\alpha,p}$: on a compact $K \subset \mathbb{R} \setminus \{0\}$ the multiplier

$$\frac{(1+\xi^2)^{\alpha/2}}{(\xi^2+\zeta^2)^{\alpha/2}}$$

and its differentiated Mihlin bounds are uniform for $\zeta \in K$. Hence $B_\zeta h = f - (V_0 - ic\zeta)h$ gives the asserted domain and its graph-norm estimate. Compact embedding on a bounded interval makes χT_ζ^{-1} compact on L^p. Therefore

$$A_\zeta = (I - \chi T_\zeta^{-1})T_\zeta$$

is Fredholm of index zero.

Suppose $A_\zeta h = 0$, with $h \in H^{\alpha,p}$ and $p > 2$. Hölder's inequality gives $L^p(\mathbb{R}) \subset \mathcal{L}_\alpha(\mathbb{R})$, and $B_\zeta h = (ic\zeta - V)h \in L^p$. Thus the distributional complex Kato inequality for B_ζ, proved after (2.5), applies directly. The imaginary term has zero real pairing with $\operatorname{sgn}_0(\overline{h})$, so

$$(B_\zeta + \mu)|h| \le (\mu - V)|h| \le f_0, \qquad f_0 = C\mathbf{1}_I|h|, \tag{6.5}$$

where I is a fixed bounded interval containing the set where $V < 2\mu$.

Put $S_t = e^{-t(B_\zeta+\mu)}$ and mollify (6.5) with a nonnegative compactly supported smooth approximate identity. Then $v_\varepsilon = |h| * \rho_\varepsilon$ belongs to $H^{\alpha,p}$ and satisfies $(B_\zeta + \mu)v_\varepsilon \le f_0 * \rho_\varepsilon$. The generator identity and positivity of S_t give

$$\begin{aligned} v_\varepsilon &= S_T v_\varepsilon + \int_0^T S_t(B_\zeta + \mu)v_\varepsilon \, dt \\ &\le S_T v_\varepsilon + \int_0^T S_t(f_0 * \rho_\varepsilon) \, dt. \end{aligned}$$

Pass to the limit first as $\varepsilon \downarrow 0$ in L^p, and then as $T \to \infty$. Since $\|S_T\|_{L^p \to L^p} \le e^{-T(\mu+|\zeta|^\alpha)}$, and the nonnegative cone of L^p is closed, this yields

$$|h| \le (B_\zeta + \mu)^{-1} f_0 \quad \text{almost everywhere.}$$

Now $f_0 \in L^1 \cap L^p$, since it is supported on I. The positive resolvent kernel has mass $(\mu+|\zeta|^\alpha)^{-1}$, so the right side belongs to L^1. Consequently $h \in L^1 \cap L^p \subset L^2$. Its equation then implies $h \in H^{\alpha,2}$. Pairing it with h in L^2 and taking imaginary parts yields

$$-c\zeta \|h\|_2^2 = 0.$$

Since $c\zeta \neq 0$, $h = 0$. Fredholm index zero gives surjectivity. Finally, the derivatives in ζ of the symbol of B_ζ are bounded multipliers from $H^{\alpha,p}$ to L^p, uniformly on compact sets avoiding zero. The inverse identity consequently gives smooth operator-norm dependence of A_ζ^{-1} on those sets. □

6.3. **Elimination of the time-dependent kernel.**

Theorem 6.3 (Homogeneous linearized rigidity)**.** *Let $0 < \alpha \le 1$, $c > 0$, and $V(x) = F''(g(x))$. Suppose that a real-valued $W \in \mathcal{L}_\alpha(\mathbb{R}^2)$ is locally $C^{1,\theta}$ for some $\theta > 0$ and solves*

$$(L_2 - c\partial_y + V(x))W = 0, \qquad \sup_x |W(x,y)| \le C(1+|y|). \tag{6.6}$$

Then $W = Aq$ for a constant A. The conclusion holds with the appropriate translated or reflected layer and its derivative. If V is instead a positive constant, $W \equiv 0$.

Proof. Transverse localization after frequency filtering. Let k be a Schwartz function with $\widehat{k} \in C_c^\infty(\mathbb{R})$ supported away from zero, and set $Y = k *_y W$. The temporal convolution preserves the tail class: translating its defining weight changes it by at most a polynomial factor, and k has moments of every order. It also preserves the linear temporal growth. Since V is independent of y, the filtered equation remains homogeneous. Convolution commutes with that equation in distributions: translate a compact test, use the polynomial weight inequality, and integrate against the Schwartz kernel. The convolution is continuous by dominated convergence and the growth bound. Choose $\mu > 0$ with $V \ge 2\mu$ outside a fixed compact interval. The distributional complex Kato inequality (2.5) applies directly to $(L_2 - c\partial_y)Y = -VY$, even when k is complex, and gives

$$(L_2 - c\partial_y + \mu)|Y| \le (\mu - V)|Y| \le C\eta(x)(1+|y|)$$

for a fixed $\eta \in C_c^\infty(\mathbb{R})$ with $\eta \ge 0$. Proposition 6.1 implies

$$|Y(x,y)| \le C\big((1+|y|)\langle x\rangle^{-1-\alpha} + \langle x\rangle^{-\alpha}\big). \tag{6.7}$$

Choose $p > \max\{2, 1/\alpha\}$. Thus Y is an L^p_x-valued tempered distribution in y with at most linear norm growth.

The operator pencil. Fix a bounded open frequency interval K with $\overline{K} \subset \mathbb{R}\setminus\{0\}$. Use bilinear duality throughout this distributional argument. The Banach-valued distribution operations can be defined explicitly. If $\psi \in C_c^\infty(K; L^{p'}(\mathbb{R}))$, where $p' = p/(p-1)$, set

$$\langle \widehat{Y}, \psi \rangle = \int_{\mathbb{R}} \left\langle Y(\cdot, y), \int_K e^{-iy\zeta} \psi(\zeta)\, d\zeta \right\rangle dy.$$

Repeated integration by parts in ζ makes the inner function rapidly decreasing in the $L^{p'}$ norm, so the linear L^p growth of Y makes this pairing absolutely convergent. A smooth operator family $M(\zeta) : E \to F$ acts by transposition, $\langle M\widehat{Y}, \psi\rangle = \langle \widehat{Y}, M(\zeta)^{\mathsf{T}}\psi(\zeta)\rangle$; here M^{T} is the transpose for bilinear duality. Smoothness and uniform operator bounds on $\overline{K}$ preserve the required test space. Sobolev-valued products are defined by this same transposition rule. In particular these products are associative, so applying an inverse family to an operator identity is legitimate. We justify the Fourier identity in the stated Banach spaces. For $\Phi \in \mathcal{S}(\mathbb{R}^2)$, symmetric second differences, split at $|H| = \langle X\rangle/2$, give

$$\sup_X \langle X\rangle^{2+\alpha} |L_2\Phi(X)| \le C\, p_N(\Phi)$$

for a fixed Schwartz seminorm p_N. Thus the equation for Y extends from compact smooth tests to all Schwartz tests by cutoff approximation and $Y \in \mathcal{L}_\alpha$. For $\Psi \in C_c^\infty(K;\mathcal{S}(\mathbb{R}_x))$, set $\Phi(x,y) = \int_K e^{-iy\zeta}\Psi(x,\zeta)\,d\zeta$. The multiplier $(\xi^2+\zeta^2)^{\alpha/2}$ is smooth on the temporal support, and hence

$$L_2\Phi(x,y) = \int_K e^{-iy\zeta} B_\zeta \Psi(x,\zeta)\,d\zeta, \qquad B_\zeta = (-\partial_{xx}+\zeta^2)^{\alpha/2}.$$

Testing the equation against Φ therefore gives $A_\zeta \widehat{Y} = 0$ first for these tests, where $A_\zeta = B_\zeta + V - ic\zeta$. On every fixed compact subset of K, three integrations by parts in ζ bound its pairing by a constant times

$$\max_{0\le j\le 3} \sup_\zeta \|\partial_\zeta^j \Psi(\zeta)\|_{H^{\alpha,p'}}.$$

Here we use the linear L^p growth of Y and the uniform multiplier bounds $\partial_\zeta^j B_\zeta : H^{\alpha,p'} \to L^{p'}$. Finite sums of scalar compact smooth functions of ζ times compact smooth functions of x are dense in $C_c^\infty(K;H^{\alpha,p'})$ on fixed compact temporal supports; this follows by partition-of-unity and Taylor approximation in ζ, followed by density of $C_c^\infty(\mathbb{R})$ in $H^{\alpha,p'}$ for the finitely many coefficients. The approximation holds for every fixed finite collection of temporal derivatives. The identity consequently holds in $\mathcal{D}'(K;H^{-\alpha,p})$.

To verify the operator domain, the equation gives

$$B_\zeta \widehat{Y} = (ic\zeta - V)\widehat{Y}.$$

Its right side belongs to $\mathcal{D}'(K;L^p)$. The smooth elliptic inverse B_ζ^{-1} is defined both from $H^{-\alpha,p}$ to L^p and from L^p to $H^{\alpha,p}$, uniformly on K. Applying it to the equality therefore gives $\widehat{Y} \in \mathcal{D}'(K;H^{\alpha,p})$. Consequently $A_\zeta \widehat{Y}$ is L^p-valued; its vanishing in $H^{-\alpha,p}$ implies its vanishing in L^p by the injective embedding. The smooth family A_ζ^{-1} from Lemma 6.2 may now multiply the vector-valued distribution, and gives $\widehat{Y} = 0$ on K. Since K was arbitrary and the temporal Fourier support of Y is compact and disjoint from zero, $Y = 0$. Every such filter annihilates W, so its partial temporal Fourier transform is supported on $\{\zeta = 0\}$. To make the resulting polynomial statement precise without imposing any integrability in x, fix $\varphi \in C_c^\infty(\mathbb{R})$ and consider

$$f_\varphi(y) = \int_{\mathbb{R}} W(x,y)\varphi(x)\,dx.$$

This is a continuous scalar function with $|f_\varphi(y)| \le C\|\varphi\|_1(1+|y|)$, and its Fourier transform is supported at zero. The structure theorem for distributions supported at one point makes f_φ a polynomial; the displayed growth bound makes its degree at most one. Its coefficients are its value at 0 and the difference between its values at 1 and 0. Hence, with $A(x) = W(x,0)$ and $B(x) = W(x,1) - W(x,0)$, one has

$$\int_{\mathbb{R}} [W(x,y) - A(x) - yB(x)]\varphi(x)\,dx = 0 \quad \text{for every } y \in \mathbb{R}.$$

Since W is continuous and φ is arbitrary, this proves

$$W(x,y) = A(x) + yB(x). \tag{6.8}$$

Both coefficients are bounded and locally $C^{1,\theta}$, as follows directly from their definitions.

The affine term lies in the ordinary fractional domain. Since A is bounded, the tail-class membership of W gives

$$\int_{\mathbb{R}} |B(x)|\langle x\rangle^{-\alpha}\,dx < \infty. \tag{6.9}$$

Indeed, integrate $|yB(x)|$ against $(1+x^2+y^2)^{-1-\alpha/2}$ and use

$$\int_{\mathbb{R}} |y|(1+x^2+y^2)^{-1-\alpha/2}\,dy = C_\alpha\langle x\rangle^{-\alpha}.$$

In the symmetric second difference for $L_2(yB(x))$, the term not equal to y times a horizontal second difference is $-r[B(x+h)-B(x-h)]$. It is $O(|(h,r)|^2)$ near the origin. Its absolute far-field integral is finite by (6.9), because integration of $|r|(h^2+r^2)^{-1-\alpha/2}$ in r gives $C_\alpha|h|^{-\alpha}$

away from $h = 0$, and boundedness handles bounded h with large r. This term is odd in r and integrates to zero. Hence

$$L_2(A + yB) = L_1A + yL_1B.$$

Substituting into (6.6) gives

$$(L_1 + V)B = 0, \qquad (L_1 + V)A = cB.$$

Lemma 2.5 gives $B = \sigma q$.

Compatibility. Testing the equation for A against q must be justified even though A is merely bounded. Both q and $L_1q = -Vq$ belong to L^1. There are $q_j \in C_c^\infty(\mathbb{R})$ with

$$q_j \to q, \qquad L_1q_j \to L_1q \quad \text{in } L^1.$$

Indeed, $q_\varepsilon = q * \rho_\varepsilon$ belongs to $W^{2,1}$, and convolution commutes with L_1, so

$$q_\varepsilon \to q, \qquad L_1q_\varepsilon = (L_1q) * \rho_\varepsilon \to L_1q \quad \text{in } L^1.$$

For fixed ε, choose $\chi \in C_c^\infty(\mathbb{R})$ equal to one near zero. Then $\chi(\cdot/R)q_\varepsilon \to q_\varepsilon$ in $W^{2,1}$ as $R \to \infty$. Symmetric second differences give $\|L_1h\|_1 \le C(\|h\|_1 + \|h''\|_1)$, so this cutoff approximation also converges in the L^1 graph norm of L_1. A diagonal sequence has the stated properties. Therefore

$$c\sigma \int qq_j = \int A(L_1 + V)q_j \longrightarrow 0.$$

It follows that $c\sigma M = 0$, so $B = 0$. The bounded stationary-kernel lemma gives $A \in \mathbb{R}q$.

For a positive constant potential $V = v_0$, Kato's inequality gives $(L_2 - c\partial_y + v_0)|W| \le 0$. Lemma 2.1 directly implies $W = 0$. Spatial translation and reflection prove the remaining versions of the theorem. □

7. Frozen reflections and normalized compactness

7.1. **Orthogonality coordinates.** We now use only consequences already proved from the original assumptions. Let $a_1^0 = \ell$ and $a_2^0 = r$ be the zero centers in Theorem 1.2.

Lemma 7.1 (Orthogonality centers)**.** *At sufficiently large heights there are C^2 centers $a_1 < a_2$, unique in fixed small neighborhoods of a_1^0, a_2^0, for which*

$$\int_{\mathbb{R}} vZ_i\,dx = 0 \quad (i = 1, 2), \qquad \begin{cases} v = u - g(x - a_1) - g(a_2 - x) + 1, \\ Z_1 = q(x - a_1), \quad Z_2 = q(a_2 - x). \end{cases} \tag{7.1}$$

They satisfy $a_i - a_i^0 \to 0$ and $a_i' \to 0$.

Proof. For fixed y let $\mathcal{F}_i(A_1, A_2, y)$ be the two left sides in (7.1). With $D = A_2 - A_1$, put

$$\mathcal{J}(D) = \int_{\mathbb{R}} g(D - t)q(t)\,dt.$$

The identities $\int gq = \frac{1}{2}[g^2]_{-\infty}^{+\infty} = 0$ and $\int q = 2$ give the useful exact formulas

$$\begin{aligned} \mathcal{F}_1(A_1, A_2, y) &= \int_{\mathbb{R}} u(A_1 + t, y)q(t)\,dt - \mathcal{J}(D) + 2, \\ \mathcal{F}_2(A_1, A_2, y) &= \int_{\mathbb{R}} u(A_2 - t, y)q(t)\,dt - \mathcal{J}(D) + 2. \end{aligned} \tag{7.2}$$

All derivatives of these expressions through order two exist and are continuous: differentiate u or g, whose derivatives through order two are bounded, against $q \in L^1$. In particular there is no need to assume $q'' \in L^1$.

At the zero-center pair $A^0(y) = (\ell(y), r(y))$, the vector $\mathcal{F}(A^0(y), y)$ tends to zero by (1.12). Write

$$K(D) = \mathcal{J}'(D) = \int_{\mathbb{R}} q(D - t)q(t)\,dt \longrightarrow 0 \quad (D \to \infty).$$

The coordinate derivative is exactly

$$D_A\mathcal{F} = \begin{pmatrix} \displaystyle\int u_x(A_1+t,y)q(t)\,dt + K(D) & -K(D) \\ K(D) & \displaystyle\int u_x(A_2-t,y)q(t)\,dt - K(D) \end{pmatrix}.$$

At $A^0(y)$ this tends to $\mathsf{D} = \operatorname{diag}(M,-M)$. The convergence needed on a whole neighborhood is uniform: if $|A - A^0(y)| \le \delta$, the global bound on u_{xx} changes the two integrals by at most $C\delta$, and the overlap $K(D)$ tends uniformly to zero there as $r - \ell \to \infty$. First choose a fixed small $\delta > 0$, then increase Y, so that

$$\|\mathsf{D}^{-1}D_A\mathcal{F} - I\| \le \tfrac{1}{2} \quad \text{on } |A - A^0(y)| \le \delta, \quad y \ge Y.$$

The map $A \mapsto A - \mathsf{D}^{-1}\mathcal{F}(A,y)$ is a contraction on that ball and maps it into itself once $|\mathsf{D}^{-1}\mathcal{F}(A^0(y),y)| \le \delta/2$. Its unique fixed point satisfies

$$|a(y) - A^0(y)| \le 2|\mathsf{D}^{-1}\mathcal{F}(A^0(y),y)| \to 0.$$

The ordinary implicit-function theorem and the joint C^2 regularity of (7.2) make these centers C^2 in y; uniqueness makes the local parameterizations agree.

Since $|a_i(y)| \to \infty$, (1.10) gives $u_y(a_1(y)+t,y), u_y(a_2(y)-t,y) \to 0$ for every fixed t. Dominated convergence against $q(t)$, using $q \in L^1$ and the global derivative bound, yields $\partial_y\mathcal{F}(a(y),y) \to 0$. Differentiating $\mathcal{F}(a(y),y) = 0$ and using the uniformly bounded inverse of $D_A\mathcal{F}$ therefore proves $a_i' \to 0$. □

Set

$$d = a_2 - a_1 \longrightarrow \infty, \qquad m = \frac{a_1+a_2}{2}, \qquad d', m' \longrightarrow 0. \tag{7.3}$$

Choose $Y > 1$ after which the coordinates are defined, and extend m constantly below Y whenever it occurs in a global growth estimate. The uniform profile approximation remains valid with these centers, since they differ from the zero centers by $o(1)$.

In centered coordinates define

$$\widetilde{u}(x,y) = u(x + m(y), y),$$
$$e(x,y) = \tfrac{1}{2}[\widetilde{u}(x,y) + \widetilde{u}(-x,y)], \qquad o(x,y) = \tfrac{1}{2}[\widetilde{u}(x,y) - \widetilde{u}(-x,y)].$$

Then e is even and o is odd. If

$$p_d(x) = g(x + d/2) + g(d/2 - x) - 1, \qquad \psi_d = (p_d)_x,$$

then (7.1) implies

$$\int o(x,y)q(x+d/2)\,dx = \int o(x,y)q(d/2-x)\,dx = 0, \qquad \int o\psi_d = 0. \tag{7.4}$$

Indeed, reflection interchanges the two tests in (7.1), while p_d is even. The centered error $\widetilde{u} - p_d$ and its reflection are therefore orthogonal to both tests, as is their odd part o. Since $\psi_d = q(x+d/2) - q(d/2-x)$, the last identity follows. The evenness of p_d also gives $\|o(\cdot,y)\|_\infty \le \|\widetilde{u}(\cdot,y) - p_d\|_\infty \to 0$. Together with $m' \to 0$, this yields

$$h(y) := \|o(\cdot,y)\|_\infty + |m'(y)| \longrightarrow 0, \qquad \sup_{y \ge Y} \|e_x(\cdot,y)\|_1 < \infty. \tag{7.5}$$

For $t, y \ge Y$, the global first-derivative bounds give

$$\|o(\cdot,y) - o(\cdot,t)\|_\infty \le C(|y-t| + |m(y) - m(t)|).$$

Since m' is continuous, h is continuous. Also $\|e_x(\cdot,y)\|_1 \le \|u_x(\cdot,y)\|_1$, which explains the uniform bound in (7.5). We will prove that h is integrable.

For an arbitrary fixed $b \in \mathbb{R}$ put

$$w_b(x,y) = \tfrac{1}{2}[u(x,y) - u(2b-x,y)],$$
$$B_b(x,y) = \int_0^1 F''(u(2b-x,y) + \theta[u(x,y) - u(2b-x,y)])\,d\theta.$$

Then

$$(L_2 - c\partial_y + B_b)w_b = 0. \tag{7.6}$$

7.2. **Localization of the exact reflection equation.**

Lemma 7.2 (The source width depends on the center, not the separation)**.** *There exist $\mu > 0$, $C < \infty$, and measurable sets $S_b \subset \mathbb{R}^2$ such that*

$$(L_2 - c\partial_y + \mu)|w_b| \le C|w_b|\mathbf{1}_{S_b}, \tag{7.7}$$

$$|S_b(y)| \le C(1 + |m(y) - b|) \quad (y \in \mathbb{R}), \tag{7.8}$$

where $S_b(y)$ is the horizontal section. The constants are independent of b. Moreover $S_b(y)$ is empty for all sufficiently negative y, uniformly in b.

Proof. Choose fixed interval neighborhoods of the two wells on which $F'' \ge 2\mu > 0$. Uniform phase geometry gives $W < \infty$ such that, at high heights, $u(x, y)$ lies in the chosen neighborhood of 1 when $a_1 + W < x < a_2 - W$, and in that of -1 when $x < a_1 - W$ or $x > a_2 + W$. Write $a_1^b = 2b - a_2$, $a_2^b = 2b - a_1$ for the reflected interfaces, and set

$$J_i(y) = [\min\{a_i, a_i^b\} - W, \max\{a_i, a_i^b\} + W], \qquad S_b(y) = J_1(y) \cup J_2(y).$$

If $x \notin S_b(y)$, it lies on the same side of each matched pair of endpoints and is more than W from all four endpoints. It belongs either to both positive-phase intervals or to neither. Thus the original and reflected values lie in the same well interval, and $B_b \ge 2\mu$. This reasoning remains valid when the joining intervals overlap. Since $|a_i^b - a_i| = 2|m(y) - b|$,

$$|S_b(y)| \le 4W + 4|m(y) - b|,$$

which proves (7.8) at high heights.

Uniform convergence to -1 below a fixed height makes S_b empty there. In the remaining bounded height band, uniform lateral clearing places all values of u outside the chosen neighborhood of -1 in a fixed horizontal compact interval. The union of that interval and its reflection has uniformly bounded length, independently of b, and supplies S_b in the band. The right side of (7.8) is at least a fixed positive constant, so one uniform C suffices. These interval constructions define a measurable set S_b. Finally Kato's inequality and (7.6) give (7.7), since B_b is uniformly bounded and is at least 2μ outside S_b. □

Lemma 7.3 (Small source mass in a controlled window)**.** *Let $b = m(y_*)$ and suppose $\sup_{|t|\le T} h(y_* + t) \le 2\varepsilon$, where the window lies above Y and $0 < \varepsilon \le 1$. Then*

$$\|w_b(\cdot, y_* + t)\|_\infty \le C\varepsilon(1 + |t|), \tag{7.9}$$

$$\int_{S_b(y_*+t)} |w_b(x, y_* + t)|\, dx \le C\varepsilon(1 + |t|) \qquad (|t| \le T). \tag{7.10}$$

Proof. Put $\delta = m(y_* + t) - b$, so $|\delta| \le 2\varepsilon|t|$. At that height,

$$w_b(b + x, y_* + t) = \tfrac{1}{2}\{e(x - \delta) - e(x + \delta) + o(x - \delta) + o(x + \delta)\}.$$

The global derivative bound proves (7.9). The fundamental theorem of calculus and Tonelli's theorem, using $\sup_{y\ge Y} \|e_x(\cdot, y)\|_1 < \infty$, give

$$\|e(\cdot - \delta) - e(\cdot + \delta)\|_1 \le 2|\delta|\|e_x\|_1 \le C|\delta|.$$

Under the change of variables $z = b + x$, the source integral is over $S_b(y_* + t) - b$ in the x variable. Integrate the odd terms on this translated set and use (7.8), since its measure is $|S_b(y_* + t)|$. The resulting bound is $C|\delta| + C\varepsilon(1 + |\delta|) \le C\varepsilon(1 + |t|)$. There is no factor $d(y_* + t)$ in this estimate. □

7.3. **Weighted positive-resolvent bounds.** Let G_c be the positive kernel of $(L_2 - c\partial_y + \mu)^{-1}$:

$$G_c(X) = \int_0^\infty e^{-\mu\tau} p_\tau^{(2)}(X + c\tau e_y)\, d\tau.$$

Here $p_\tau^{(2)}$ is the fractional heat kernel. Positivity and marginalization give $\int G_c = \mu^{-1}$, and

$$G_c(X) \le C|X|^{-2-\alpha} \qquad (|X| \ge 1). \tag{7.11}$$

This is the bound proved in (5.13). Positive bounded comparison in (7.7) yields

$$|w_b| \le C G_c * (|w_b| \mathbf{1}_{S_b}). \tag{7.12}$$

Only this analytic semigroup formula and its positive kernel are used.

Put $\rho(X) = (1 + |X|^2)^{-1-\alpha/2}$. The kernel satisfies

$$\int \rho(X) G_c(X - Z)\, dX \le C\rho(Z). \tag{7.13}$$

For $|Z| \le 2$, use $\rho \le 1$, the kernel mass, and the positive lower bound for $\rho(Z)$. For $|Z| > 2$, split according to whether $|X - Z| \le |Z|/2$. On this part $\rho(X) \le C\rho(Z)$, so the kernel mass applies. On the complementary part, (7.11) gives $G_c(X - Z) \le C\rho(Z)$; integrate against $\rho \in L^1$ to obtain (7.13). It follows that convolution with G_c preserves uniformly bounded, tight weighted $L^1(\rho)$ families. Here tightness means $\lim_{R\to\infty} \sup_j \int_{|Z|>R} \rho(Z)|f_j(Z)|\, dZ = 0$. For an explicit verification, set

$$\Xi_R(Z) = \int_{|X|>R} \rho(X) G_c(X - Z)\, dX.$$

Equation (7.13) gives $\Xi_R(Z) \le C\rho(Z)$. For every fixed R_0, the ratio $\Xi_R(Z)/\rho(Z)$ tends uniformly to zero on $|Z| \le R_0$: when $R > 2R_0 + 2$, use the far-field bound for G_c and then the integrability of $|X|^{-4-2\alpha}$. Positivity and Tonelli now yield

$$\begin{aligned}\int_{|X|>R} \rho(X)(G_c * |f_j|)(X)\, dX \le C \int_{|Z|>R_0} \rho(Z)|f_j(Z)|\, dZ \\ + \sup_{|Z|\le R_0} \frac{\Xi_R(Z)}{\rho(Z)} \int \rho(Z)|f_j(Z)|\, dZ.\end{aligned}$$

First choose R_0 using tightness, and then R using the compact uniform limit. This proves the claimed preservation property, including for kernels that are not even because of the drift.

Lemma 7.4 (Tail-tight normalized reflections)**.** *Assume $|m(y)| \le C(1 + y)^p$ for $y \ge Y$, with $0 \le p \le 1$. Let $y_j \to \infty$, $b_j = m(y_j)$, $\varepsilon_j = h(y_j) > 0$, and $T_j \to \infty$ satisfy*

$$T_j = o(y_j), \qquad \sup_{|t|\le T_j} h(y_j + t) \le 2\varepsilon_j, \qquad \varepsilon_j^{-1}(1 + y_j)^p T_j^{-1-\alpha} \to 0. \tag{7.14}$$

For arbitrary horizontal observation points x_j, the functions

$$W_j(x, t) = \frac{w_{b_j}(x_j + x, y_j + t)}{\varepsilon_j}$$

have uniformly bounded and tight $L^1(\rho)$ tails, uniformly in the horizontal observation points. Moreover, for every fixed $0 < T < \infty$,

$$|W_j(x, t)| \le C(1 + |t|) \qquad (x \in \mathbb{R},\ |t| \le T)$$

for all sufficiently large j, with C independent of T and x_j.

Proof. Normalize and translate the nonnegative source in (7.12):

$$f_j(x, t) = \varepsilon_j^{-1} |w_{b_j}(x_j + x, y_j + t)| \mathbf{1}_{S_{b_j}}(x_j + x, y_j + t).$$

Since $h(y_j) \to 0$ and $T_j = o(y_j)$, for all sufficiently large j one has $0 < \varepsilon_j \le 1$, $T_j \ge 2$, and $y_j - T_j \ge Y$. Inside these windows Lemma 7.3 gives $\|f_j(\cdot, t)\|_1 \le C(1 + |t|)$, independently of

x_j. For $R \geq 2$, split the complement of the disk into $|t| > R/2$ and $|t| \leq R/2$, $|x| > \sqrt{3}R/2$. Its weighted mass from inside the window is at most

$$C \int_{|t|>R/2} (1+|t|)^{-2-\alpha}(1+|t|)\,dt + CR^{-2-\alpha} \int_{-R/2}^{R/2} (1+|t|)\,dt \leq CR^{-\alpha}.$$

Its entire weighted norm is bounded by $C \int_{\mathbb{R}} \langle t \rangle^{-2-\alpha}(1+|t|)\,dt < \infty$.

Outside the window, $|w_b| \leq 1$ and the source-width bound give

$$\|f_j(\cdot,t)\|_1 \leq \frac{C}{\varepsilon_j}((1+y_j)^p + (1+|t|)^p).$$

The constant extension of m below Y justifies the same estimate at negative heights. Since $p < 1+\alpha$ and $T_j = o(y_j)$, the weighted mass of this part is at most

$$\frac{C}{\varepsilon_j}((1+y_j)^p T_j^{-1-\alpha} + T_j^{p-1-\alpha}) \leq \frac{C}{\varepsilon_j}(1+y_j)^p T_j^{-1-\alpha} \longrightarrow 0.$$

The estimates are independent of x_j. They prove

$$\lim_{R\to\infty} \limsup_{j\to\infty} \int_{|X|>R} \rho(X) f_j(X)\,dX = 0$$

and a uniform weighted norm bound for all large j. For each of the finitely many remaining indices, $0 \leq f_j \leq \varepsilon_j^{-1}$, so

$$\int_{|X|>R} \rho(X) f_j(X)\,dX \leq \varepsilon_j^{-1} \int_{|X|>R} \rho(X)\,dX \longrightarrow 0.$$

This bound is independent of x_j and uses only $\rho \in L^1$. Combining these finite-index bounds with the displayed iterated limit gives uniform tightness for the whole sequence. Equations (7.12) and (7.13), with the preservation property just proved, give the same conclusion for W_j. Its pointwise bound on every fixed temporal interval, once T_j contains that interval, is (7.9). □

7.4. **Local compactness and a uniform derivative estimate.** The normalized functions in Lemma 7.4 solve

$$(L_2 - c\partial_t + B_j)W_j = 0,$$

where B_j are the translated reflection coefficients. These coefficients have uniform C^1 bounds: differentiating the integral defining B_b uses only boundedness of F''' on $[-1,1]$ and the global first derivatives of u. On every fixed ball, the pointwise estimate of Lemma 7.3 applies for all sufficiently large j, because $T_j \to \infty$. Together with the uniform weighted norm bound of Lemma 7.4, it supplies the right side of (2.7) uniformly. Lemma 2.4 therefore gives uniform $C^{1,\theta}_{\mathrm{loc}}$ bounds and subsequential convergence in C^1_{loc}. By the uniform C^1 coefficient bounds, a further subsequence has $B_j \to B_\infty$ locally uniformly. Tightness then passes the fractional operator and the equation to the limit, including at $\alpha = 1$, so $(L_2 - c\partial_t + B_\infty)W = 0$. Fatou's lemma puts every limit in $\mathcal{L}_\alpha$. The pointwise window estimate passes to the limit at every (x,t) and gives

$$\sup_x |W(x,t)| \leq C(1+|t|).$$

The coefficient bounds, local pointwise bounds, and weighted norms are independent of the horizontal observation point. Thus applying (2.7) on a fixed ball centered at (x, y_j), and then taking the supremum over x, gives, for all sufficiently large j,

$$\sup_x \frac{|\partial_y w_{b_j}(x,y_j)|}{\varepsilon_j} \leq C. \tag{7.15}$$

This derivative estimate will be needed when differentiating the orthogonality constraint; convergence of the normalized shape alone would not control the midpoint velocity.

8. The growth-to-decay bootstrap and the limiting midpoint

8.1. Improvement of a polynomial growth bound.

Proposition 8.1 (Growth-to-decay improvement)**.** *Suppose, for some $p \in [0,1]$, that $|m(y)| \le C(1+y)^p$ at large heights. Then*

$$h(y) = \|o(\cdot,y)\|_\infty + |m'(y)| \le C_p(1+y)^{-(1+\alpha-p)}. \tag{8.1}$$

Proof. Write $\beta = 1+\alpha-p > 0$. Suppose that $h(y)(1+y)^\beta$ is unbounded. Choose starting points $Y_j \to \infty$ with this product tending to infinity, and choose $k_j \to \infty$ so slowly that

$$k_j\big[h(Y_j)(1+Y_j)^\beta\big]^{-1/(1+\alpha)} \to 0.$$

For completeness, implement the doubling construction with the fixed factor $A_j = (1+Y_j)^p$. At a current point z use the radius $r(z) = k_j(A_j/h(z))^{1/(1+\alpha)}$. Start at $z = Y_j$. If $h \le 2h(z)$ throughout $[z-r(z), z+r(z)]$, stop. Otherwise move to a point in that interval where $h > 2h(z)$ and repeat. At successive moves the radii decrease by at least the factor $2^{-1/(1+\alpha)}$. Therefore the total travel, including the last window radius, is at most

$$C_\alpha r(Y_j), \qquad \frac{r(Y_j)}{1+Y_j} = k_j\big[h(Y_j)(1+Y_j)^{1+\alpha-p}\big]^{-1/(1+\alpha)} \to 0.$$

For all large j, every window stays in $[Y_j/2, 3Y_j/2]$ and hence above Y. Infinite doubling would make the continuous function h unbounded on this fixed compact interval, so the procedure stops. Denote its endpoint by y_j, and set $\varepsilon_j = h(y_j) > 0$. Then $y_j \sim Y_j$, $\varepsilon_j \to 0$, and

$$T_j = k_j(A_j/\varepsilon_j)^{1/(1+\alpha)} \le r(Y_j) = o(y_j),$$
$$\sup_{|t|\le T_j} h(y_j+t) \le 2\varepsilon_j.$$

Also $T_j \to \infty$, since $A_j \ge 1$, $k_j \to \infty$, and $\varepsilon_j \to 0$. Finally,

$$\varepsilon_j^{-1}(1+y_j)^p T_j^{-1-\alpha} = \frac{(1+y_j)^p}{k_j^{1+\alpha}(1+Y_j)^p} \to 0.$$

Thus every part of (7.14) is verified.

Set $b_j = m(y_j)$ and define the normalized profiles with frozen centers

$$U_{i,j}(X,t) = \frac{w_{b_j}(a_i(y_j)+X, y_j+t)}{\varepsilon_j} \qquad (i=1,2).$$

Lemma 7.4 and the local compactness from Section 7 apply to these profiles. For each fixed $T > 0$,

$$\sup_{|t|\le T} |a_i(y_j+t) - a_i(y_j)| \le T \sup_{s \ge y_j - T} |a_i'(s)| \longrightarrow 0 \qquad (i=1,2).$$

In the left-core coordinates $x = a_1(y_j) + X$, the identity $2b_j - a_1(y_j) = a_2(y_j)$ shows that the two functions entering the frozen reflection coefficient converge locally to the same layer $g(X)$. Indeed, apply (1.12) at $y_j + t$, use the displayed increment estimate, and let $d(y_j+t) \to \infty$. The global second derivative bounds promote this convergence to C^1_{loc}. Similarly, the right-core backgrounds converge to $g(-X)$. Consequently every normalized core limit satisfies the corresponding homogeneous layer equation. Its tail membership and linear temporal growth were established above, so Theorem 6.3 makes it a constant multiple of the corresponding layer derivative.

At $t = 0$, (7.4) eliminates this multiple. For the left core, the identity is exactly

$$\int_{\mathbb{R}} \frac{w_{b_j}(a_1(y_j)+X, y_j)}{\varepsilon_j} q(X)\, dX = 0.$$

The absolute value of the integrand is at most $q(X)$, because $w_{b_j}(b_j+X, y_j) = o(X, y_j)$ and $\|o(\cdot, y_j)\|_\infty \le \varepsilon_j$. Dominated convergence gives $AM = 0$ for a limit Aq. The right-core test is

$q(-X)$ and has the same conclusion. Every convergent subsequence thus has zero limit. Local compactness therefore gives $U_{i,j} \to 0$ in $C^1_{\mathrm{loc}}(\mathbb{R}^2)$ for both full sequences.

We claim that this also gives the uniform statement

$$\|o(\cdot, y_j)\|_\infty/\varepsilon_j \longrightarrow 0. \tag{8.2}$$

If (8.2) failed, choose physical horizontal coordinates x_j with $|w_{b_j}(x_j, y_j)|/\varepsilon_j \geq \delta > 0$. After a subsequence, either x_j stays a bounded distance from one of $a_1(y_j), a_2(y_j)$, or its distances from both tend to infinity. The first case contradicts the normalized core convergence. In the second case set $x_j^* = 2b_j - x_j$. Since $2b_j = a_1(y_j) + a_2(y_j)$, reflection swaps the two distances and preserves membership in the interval $(a_1(y_j), a_2(y_j))$. Choose a subsequence on which x_j is always inside that interval or always outside it, and let $\sigma = 1$ or -1, respectively. For bounded X, t, the points $x_j + X$ and $x_j^* - X$ remain arbitrarily far from both interfaces at height $y_j + t$, because the interface displacements $a_i(y_j + t) - a_i(y_j)$ tend uniformly to zero on compact temporal intervals. Their interior or exterior status is therefore unchanged. The uniform two-layer approximation now gives

$$u(x_j + X, y_j + t) \longrightarrow \sigma, \qquad u(x_j^* - X, y_j + t) \longrightarrow \sigma$$

locally uniformly in (X, t). Hence the reflection coefficient tends to the positive constant $F''(\sigma)$. Lemma 7.4 and local compactness produce a limit of $w_{b_j}(x_j + X, y_j + t)/\varepsilon_j$ with absolute value at least δ at the origin, belonging to $\mathcal{L}_\alpha$ and having the required linear temporal growth. The positive-constant case of Theorem 6.3 makes this limit zero, a contradiction. This proves (8.2).

It remains to control the normalized midpoint velocity, not only the shape. Put $d_j = d(y_j)$. At the instant y_j, with the reflection axis b_j held fixed, the chain rule gives

$$\partial_y w_{b_j}(b_j + x, y_j) = o_y(x, y_j) - m'(y_j) e_x(x, y_j).$$

The differentiation of (7.4) is justified in normed spaces: (2.11) gives $q \in W^{1,1}$, so $d \mapsto \psi_d$ is a C^1 map into L^1, while $y \mapsto o(\cdot, y)$ is C^1 into L^∞ on compact temporal intervals. The latter follows from the global second derivatives of u and the C^2 regularity of m. The product rule therefore gives

$$\int \partial_y w_{b_j}(b_j + x, y_j)\psi_{d_j}(x)\,dx = -m'(y_j)\mathfrak{M}_j - d'(y_j)\int o(x, y_j)\partial_d \psi_{d_j}(x)\,dx, \tag{8.3}$$

where

$$\mathfrak{M}_j = \int e_x(x, y_j)\psi_{d_j}(x)\,dx,$$

$$\partial_d \psi_d(x) = \tfrac{1}{2}[q'(x + d/2) - q'(d/2 - x)].$$

Thus $\|\partial_d \psi_d\|_1 \leq \|q'\|_1$. Uniform C^1 profile approximation implies $\|e_x(\cdot, y_j) - \psi_{d_j}\|_\infty \to 0$, while $\|\psi_{d_j}\|_1 \leq 4$. Consequently

$$\mathfrak{M}_j = \int \psi_{d_j}^2 + o(1) = 2M - 2\int q(x + d_j/2)q(d_j/2 - x)\,dx + o(1) \longrightarrow 2M > 0.$$

Divide the left side of (8.3) by ε_j. On the neighborhoods $|x + d_j/2| \leq R$ and $|x - d_j/2| \leq R$, it tends to zero because $\partial_t U_{i,j}(\cdot, 0) \to 0$ uniformly on bounded intervals, with the physical centers $a_i(y_j)$ held fixed. On their complement both translated copies of q are in their tails. Equation (7.15) therefore bounds the absolute value of this complementary pairing by $2C\int_{|t|>R} q(t)\,dt$, uniformly in j. Sending first j and then R to infinity proves convergence of the full pairing to zero. The remaining term in (8.3), divided by ε_j, has absolute value at most $|d'(y_j)|\|q'\|_1 \to 0$. Since $\mathfrak{M}_j \to 2M > 0$, it follows that $m'(y_j)/\varepsilon_j \to 0$. Together with (8.2), this contradicts $\|o(\cdot, y_j)\|_\infty/\varepsilon_j + |m'(y_j)|/\varepsilon_j = 1$. This proves (8.1). □

8.2. **Finite iteration, including the logarithmic borderline.**

Proof of Theorem 1.3, except the identification of the axis. Initially $m' = o(1)$ gives $|m(y)| \leq C(1 + y)$, so Proposition 8.1 applies with $p = 1$. If a current growth exponent satisfies $p > \alpha$, then

$$|m'| \leq C(1 + y)^{-1-\alpha+p} \quad \Longrightarrow \quad |m(y)| \leq C(1 + y)^{p-\alpha}.$$

The exponent is reduced by α. If $p < \alpha$, the velocity is already integrable. If $p = \alpha$, the estimate first gives $|m(y)| \le C\log(2+y)$. Since $\log(2+y) \le C(1+y)^{\alpha/2}$, one more application has an integrable velocity. After finitely many steps for each fixed $\alpha > 0$,

$$m' \in L^1(Y,\infty), \qquad m(y) \longrightarrow b \in \mathbb{R}. \tag{8.4}$$

Applying Proposition 8.1 once more with $p = 0$ gives

$$|m'(y)| + \|o(\cdot, y)\|_\infty \le C(1+y)^{-1-\alpha}.$$

Integration from y to infinity gives $|m(y) - b| \le C(1+y)^{-\alpha}$, which is (1.13). Lemma 7.1 supplies the other coordinate assertions. The identification of b with the global reflection axis will be proved in Section 9. □

Remark 8.2 (Critical order and small positive orders). At $\alpha = 1$, the final estimates are $h = O(y^{-2})$ and $m - b = O(y^{-1})$. For every fixed $\alpha > 0$, the iteration is finite, but its number of steps is not bounded uniformly as $\alpha \downarrow 0$. The iteration never differentiates F to progressively higher orders. Neither a sharp separation law nor an assumed L^1 bound for a nonlinear odd forcing enters the proof.

9. Moving planes and proof of horizontal symmetry

For $\lambda \in \mathbb{R}$ put

$$\Sigma_\lambda = \{(x,y) : x < \lambda\}, \qquad X^\lambda = (2\lambda - x, y), \qquad z_\lambda(X) = u(X^\lambda) - u(X).$$

The difference is antisymmetric across $x = \lambda$ and solves

$$(L_2 - c\partial_y + B_\lambda) z_\lambda = 0, \qquad \|B_\lambda\|_\infty \le K. \tag{9.1}$$

Here $K = \max_{[-1,1]} |F''|$ and B_λ is the coefficient B_b in (7.6) with $b = \lambda$: indeed, $z_\lambda = -2w_\lambda$. In particular, this coefficient is defined even when the compared values coincide. The global regularity from Section 2 gives $z_\lambda \in C_b^{2,\gamma}$, and the uniform C^1 bounds for B_b established in Section 7 give uniform Lipschitz bounds for B_λ. Choose $\mu > 0$ so small that two values in the same sufficiently small well neighborhood have $B_\lambda \ge 2\mu$.

9.1. **Comparison at the upper end.**

Lemma 9.1 (Upper-end comparison geometry)**.** *For each $\delta > 0$ there is T_δ such that, whenever $y \ge T_\delta$, $\lambda \le m(y) - \delta$, and $x < \lambda$,*

$$z_\lambda(x,y) < 0 \quad \Longrightarrow \quad B_\lambda(x,y) \ge \mu.$$

The conclusion is uniform over all such planes, including planes tending to $-\infty$.

Proof. If the assertion failed, there would be $y_j \to \infty$ and $x_j < \lambda_j \le m(y_j) - \delta$ with $z_{\lambda_j}(x_j, y_j) < 0$ and $B_{\lambda_j}(x_j, y_j) < \mu$. Put $X_j = x_j$, $Z_j = 2\lambda_j - x_j > X_j$, and $a_i = a_i(y_j)$. Pass to subsequences so that each comparison point either stays a bounded distance from one core or has distances from both tending to infinity; every bounded oriented core coordinate may also be assumed to converge. Since

$$a_2 - X_j > \frac{a_2 - a_1}{2} + \delta \longrightarrow \infty,$$

only the left core can be near X_j. We shall use the exact identity

$$a_2 - Z_j = X_j - a_1 + 2(m(y_j) - \lambda_j) \ge X_j - a_1 + 2\delta. \tag{9.2}$$

The profile approximation (1.12), valid also with the orthogonality centers, determines the limits in the following cases.

If both comparison points stay far from both cores, each value tends to a well after a further subsequence. The same well is excluded by $B_{\lambda_j} < \mu$. The only other possibility compatible with $z_{\lambda_j} < 0$ is $u(X_j, y_j) \to 1$, $u(Z_j, y_j) \to -1$. Since X_j is on the left of the midpoint, its limit 1 forces $X_j - a_1 \to \infty$. Then (9.2) gives $a_2 - Z_j \to \infty$, while $Z_j - a_1 > X_j - a_1 \to \infty$. Thus Z_j is in the interior phase and its value tends to 1, a contradiction.

If X_j stays near a core, write $X_j - a_1 \to \xi \in \mathbb{R}$ and $t_j = \lambda_j - X_j > 0$. Pass to a subsequence for which t_j tends to zero, to a positive finite limit, or to infinity. If $t_j \to 0$, the interval $[X_j, Z_j]$ lies in a fixed left-core neighborhood. The C^1 layer convergence and $g' > 0$ give $u_x > 0$ on that interval for large j, and hence

$$z_{\lambda_j}(x_j, y_j) = \int_{X_j}^{Z_j} u_x(t, y_j)\, dt > 0.$$

If $t_j \to t \in (0, \infty)$, the limiting difference is $g(\xi + 2t) - g(\xi) > 0$. If $t_j \to \infty$, then $Z_j - a_1 \to \infty$. Put $\rho_j = a_2 - Z_j \geq X_j - a_1 + 2\delta$. After a further subsequence, either $\rho_j \to \infty$, in which case Z_j is in the interior phase and the difference tends to $1 - g(\xi) > 0$, or $\rho_j \to \rho \in \mathbb{R}$ with $\rho \geq \xi + 2\delta$. In the latter case right-core convergence gives the limiting difference $g(\rho) - g(\xi) > 0$. The lower bound on ρ_j excludes escape to the right exterior. Each subcase contradicts the negative comparison.

Finally, suppose only Z_j stays near a core. Then $X_j - a_1$ is bounded above: use $X_j < Z_j$ for the left core, and (9.2) for the right core. Since X_j stays far from the cores, $X_j - a_1 \to -\infty$, so $u(X_j, y_j) \to -1$. The value at Z_j converges to g at a finite oriented argument, which is strictly greater than -1. The difference is therefore eventually positive. These cases exhaust all possibilities and prove the assertion. □

9.2. **Antisymmetric maximum principles with tangential drift.**

Lemma 9.2 (Stable negative sets and narrow strips)**.** *Let $z \in C_b^{1,\theta}(\mathbb{R}^2)$ for some $0 < \theta < 1$ solve $(L_2 - c\partial_y + B)z = 0$ in Σ_λ and be antisymmetric about $x = \lambda$. Assume that B is bounded and uniformly continuous. Since $\alpha \leq 1$, one has $1+\theta > \alpha$, so the fractional operator is classical. If $B \geq \mu > 0$ on $\{z < 0\} \cap \Sigma_\lambda$, then $z \geq 0$ in Σ_λ. More generally, the same conclusion holds if every negative point either has $B \geq \mu$ or lies in a strip $0 < \lambda - x < \eta$, where η is sufficiently small in terms of α and $\|B\|_\infty$. If $z \geq 0$ in Σ_λ and has a zero there, then $z \equiv 0$.*

Proof. First suppose that a negative infimum over Σ_λ is attained at $X \in \Sigma_\lambda$. The minimum here is over the half-space; no minimum over the reflected half-space is required. Splitting the principal-value integral into the two half-spaces and changing variables by reflection in the second gives

$$L_2 z(X) = C_{2,s} \int_{\Sigma_\lambda} (z(X) - z(Z))\big(|X - Z|^{-2-\alpha} - |X - Z^\lambda|^{-2-\alpha}\big)\, dZ + 2C_{2,s} z(X) \int_{\Sigma_\lambda} |X - Z^\lambda|^{-2-\alpha}\, dZ.$$

At this interior minimum $\nabla z(X) = 0$, so $|z(X) - z(Z)| \leq C|X - Z|^{1+\theta}$ near X. Since $1+\theta > \alpha$, the displayed first integral is absolutely convergent; thus the principal-value splitting is justified. Its remaining terms are integrable because X has positive distance from the plane and z is bounded. The kernel difference is strictly positive. The first integral is nonpositive, since $z(X) \leq z(Z)$ on the original half-space. Dimensional marginalization gives

$$2C_{2,s} \int_{\Sigma_\lambda} |X - Z^\lambda|^{-2-\alpha}\, dZ = 2C_{1,s} \int_{\lambda - x}^{\infty} r^{-1-\alpha}\, dr = \frac{2C_{1,s}}{\alpha} (\lambda - x)^{-\alpha}.$$

Thus, with $c_\alpha = 2C_{1,s}/\alpha > 0$,

$$L_2 z(X) \leq c_\alpha z(X) (\lambda - x)^{-\alpha}. \tag{9.3}$$

At the minimum $\partial_y z(X) = 0$. If $B(X) \geq \mu$, the left side of its equation is strictly negative. If instead $\lambda - x \leq \eta$, it is again strictly negative as soon as

$$c_\alpha \eta^{-\alpha} > \|B\|_\infty.$$

This proves the attained-minimum assertion, also with the non-strict strip width that can arise in a limit.

For a nonattained negative infimum ν, choose $X_j = (x_j, y_j) \in \Sigma_\lambda$ with $z(X_j) \to \nu < 0$, and put $d_j = \lambda - x_j$. Since $z = 0$ on the plane and ∇z is bounded, d_j is bounded away

from zero. Translate both z and B by X_j. The global $C^{1,\theta}$ bound and uniform continuity of B yield subsequential local convergence of these translates in C^1 and C^0, respectively. Their uniform L^∞ bounds control the far-field fractional tails. For the near field, symmetric second differences and the common $C^{1,\theta}$ bound give a uniform bound $Cr^{1+\theta-\alpha}$ for the integral over increments of size less than r. This tends to zero, including at $\alpha = 1$; local C^1 convergence handles the intermediate annulus and the drift term. Hence the translated equations pass to the limit wherever the limiting domain is open.

If d_j stays bounded, take $d_j \to d > 0$. The limiting solution is antisymmetric about $x = d$, has its half-space minimum ν at the origin, and solves the limiting equation there. Along a further subsequence either $B(X_j) \geq \mu$ for every j, or every X_j lies in the narrow strip. In the first case the limiting coefficient at the origin is at least μ; in the second $d \leq \eta$. The attained-minimum argument gives a contradiction in either case. If $d_j \to \infty$, every fixed ball lies in the translated original half-space for large j. The limiting function has a whole-space minimum ν at the origin, and $B(X_j) \geq \mu$ because these points eventually lie outside the strip. Thus $L_2 z_\infty(0) \leq 0$, $\partial_y z_\infty(0) = 0$, and $B_\infty(0) z_\infty(0) < 0$, again contradicting the equation. This proves the weak conclusions.

Finally, at an interior zero of a nonnegative antisymmetric solution, the same half-space identity reduces to the negative integral of $z(Z)$ times the strictly positive kernel difference. The drift and potential terms vanish there. Hence that integral is zero and $z = 0$ throughout the half-space by continuity. Antisymmetry gives $z \equiv 0$ on $\mathbb{R}^2$. □

9.3. **Starting and continuing the plane.**

Proof of Theorem 1.1. First observe that any reflection axis a must equal the limit b in (8.4). Indeed, at large heights reflection must interchange the only two zeros $\ell(y), r(y)$, so $\ell(y)+r(y) = 2a$. Since $a_1 - \ell \to 0$ and $a_2 - r \to 0$, this gives $m(y) \to a$ and hence $a = b$.

Starting the plane. By (8.4), choose T large enough that $m(y) \geq b - 1$ for $y \geq T$, and increase T to exceed the height in Lemma 9.1 for $\delta = 1$. Then, for every $\lambda \leq b - 2$ and $y \geq T$, negative comparison values have $B_\lambda \geq \mu$. Uniform lateral clearing for $y \leq T$ gives A such that $u(x, y)$ is near -1 whenever $x < -A$, $y \leq T$. Choose $\lambda_0 < \min(-A, b - 2)$. For $\lambda \leq \lambda_0$, a negative comparison on this lower region makes the reflected value still smaller, so both values are near -1. Thus $B_\lambda \geq \mu$ on the entire negative set, and Lemma 9.2 gives $z_\lambda \geq 0$.

Continuation up to b. Define

$$\lambda_* = \sup\{\lambda \leq b : z_\tau \geq 0 \text{ in } \Sigma_\tau \text{ for every } \tau \leq \lambda\}.$$

The defining set is nonempty by the starting argument. For every $\tau < \lambda_*$, its definition gives $z_\tau \geq 0$ in Σ_τ. Choose $\tau_j \uparrow \lambda_*$. For each fixed $X \in \Sigma_{\lambda_*}$ one has $X \in \Sigma_{\tau_j}$ eventually, and continuity in the plane parameter gives $z_{\lambda_*}(X) \geq 0$. Thus the endpoint is admissible, including if $\lambda_* = b$.

Suppose $\lambda_* < b$. The strong antisymmetric principle gives strict positivity in Σ_{λ_*}: the alternative $z_{\lambda_*} \equiv 0$ would make λ_* a reflection axis, contrary to the initial observation.

Fix $\varepsilon_0 > 0$ so that $\lambda_* + \varepsilon_0 < b$, and put $\delta = (b - \lambda_* - \varepsilon_0)/2 > 0$. At sufficiently large heights $|m(y) - b| < \delta$, so every plane $\lambda \in [\lambda_*, \lambda_* + \varepsilon_0]$ satisfies $\lambda \leq m(y) - \delta$. Lemma 9.1 therefore gives a common upper height Y_+ above which negative values have $B_\lambda \geq \mu$. Uniform negative clearing gives a common lower height $Y_- < Y_+$, below which both comparison values are near -1. Lateral clearing for $y \leq Y_+$ gives $A < \infty$ such that a negative comparison with $x < -A$ also has $B_\lambda \geq \mu$: the unreflected value is near -1, and a negative difference makes the reflected value still smaller. Thus all negative points with $B_\lambda < \mu$ belong to

$$[-A, \lambda_* + \varepsilon_0] \times [Y_-, Y_+].$$

Choose $\eta > 0$ so small that $c_\alpha \eta^{-\alpha} > K$, and increase A so that the following compact rectangle is nonempty:

$$\mathcal{K}_\eta = [-A, \lambda_* - \eta/2] \times [Y_-, Y_+] \Subset \Sigma_{\lambda_*}.$$

Strict positivity gives $\sigma := \min_{\mathcal{K}_\eta} z_{\lambda_*} > 0$. Choose

$$0 < \varepsilon < \min\left\{\varepsilon_0, \frac{\eta}{2}, \frac{\sigma}{4(1+\|u_x\|_\infty)}\right\}.$$

For $\lambda \in [\lambda_*, \lambda_* + \varepsilon]$, $|z_\lambda - z_{\lambda_*}| \le 2\|u_x\|_\infty \varepsilon$, so $z_\lambda > 0$ on $\mathcal{K}_\eta$. If a negative point with $B_\lambda < \mu$ were outside the strip $0 < \lambda - x < \eta$, then $x \le \lambda - \eta \le \lambda_* - \eta/2$, placing it in $\mathcal{K}_\eta$, a contradiction. Every negative point therefore has a positive coefficient or lies in the prescribed narrow strip. Lemma 9.2 proves $z_\lambda \ge 0$ for every plane in this interval. This contradicts the definition of λ_*. Hence $z_\lambda \ge 0$ for all $\lambda \le b$.

To obtain comparison from the right, set $\bar{u}(x,y) = u(-x,y)$. Horizontal reflection preserves the equation and all the hypotheses; the zero branches of $\bar{u}$ are $-r(y), -\ell(y)$, so its limiting midpoint is $-b$. Applying the left-plane comparison just proved to $\bar{u}$ at $-b$ gives $z_b(x,y) \ge 0$ for $x > b$. Together with the left-plane comparison and the antisymmetry of z_b, this yields

$$u(2b - x, y) = u(x, y) \qquad \text{everywhere.}$$

For every $\lambda < b$, Lemma 9.2 makes the comparison strict in Σ_λ, since equality would give a reflection axis different from b. If $x_1 < x_2 \le b$, apply this strict comparison at $\lambda = (x_1 + x_2)/2 < b$ to obtain $u(x_2, y) > u(x_1, y)$. Differentiability and reflection now give $u_x \ge 0$ for $x < b$ and $u_x \le 0$ for $x > b$. By the differentiated equation established in Section 2, $u_x \in C_b^{1,\gamma}$ satisfies classically

$$(L_2 - c\partial_y + F''(u))u_x = 0.$$

It is antisymmetric about b, and the strict comparisons show that it is not identically zero. Since $F''(u)$ is bounded and uniformly continuous, the strong alternative in Lemma 9.2 gives $u_x > 0$ on the left; reflection gives $u_x < 0$ on the right. Uniqueness of the axis follows from the initial observation. This proves (1.5) and identifies the limit in (8.4) with that axis, completing also Theorem 1.3. □

Remark 9.3 (Exact centering after symmetry). Reflection about b maps an orthogonality pair (a_1, a_2) to $(2b - a_2, 2b - a_1)$: changing variables $x \mapsto 2b - x$ in (7.1) interchanges the two orthogonality conditions, since $u(2b - x, y) = u(x, y)$. Moreover, $\ell + r = 2b$, so the deviations of the reflected pair from (ℓ, r) are $-(a_2 - r)$ and $-(a_1 - \ell)$. They tend to zero, and both pairs therefore lie in the uniqueness neighborhood from Lemma 7.1 at large heights. That uniqueness gives $a_1 + a_2 = 2b$. Accordingly, $m \equiv b$ and $o \equiv 0$ there. The decay estimates used to reach the axis were not assumptions of this final symmetry argument.

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